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Review Article

Passive cooling of Photovoltaic panels using radiative paints and metal foam fins: A critical review of mechanisms, materials, and performance metrics

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ABSTRACT

The performance and lifespan of solar Photovoltaic (PV) panels can be drastically improved by minimizing thermal buildup, which can constrain performance. This review critically compares two trending passive cooling technologies, such as radiative paints and metal foam fins, across their mechanisms, materials, and performance metrics. Radiative paints that enable solar radiation reflection and thermal emission, and metal foam fins that increase heat dissipation by providing increased surface area, are investigated separately and in combination. We discuss the contributions of these techniques to reduced operational temperatures, improved energy yield, and enhanced PV panel longevity. The review integrates discussion of the materials used in these technologies, the difficulties in their implementation, and the need for standard test procedures. We also cover multidimensional performance tests, including thermal conductivity, cooling performance, and environmental footprint. The review also provides current research gaps and recommends future research and development directions in hybrid cooling systems, the incorporation of new materials, and large-scale experimental verification. The synthesis offers a comprehensive review of the prospects of passive cooling technologies for optimizing PV systems, with implications for both academic research and real-world applications in the solar industry.

1. Introduction

Over the past two decades, the Photovoltaic (PV) sector has experienced rapid and sustained growth, driven by continuous innovation and favorable economics. Global PV production has increased exponentially since 2002, doubling approximately every 2 years and averaging 48 % growth per annum [1]. Asia-Pacific became the world leader in PV deployment in 2018, with China accounting for 58 % of the world's installed capacity [2]. The global total capacity amounted to 591 GW by 2019, driven by rising module efficiency, falling production costs, and the growing competitiveness of solar power [3]. Forecasting indicates that solar PV systems can achieve an installed capacity of approximately 8516 GW by 2050, with an average annual growth rate of 9 % from 2019 to 2050. The capacity would account for a very significant share of global electricity generation, surpassing previous projections of 6750 TWh annually, as shown in Fig. 1 [4,5].

As solar power continues to expand, one major challenge that persists is the thermal management of PV systems. Despite significant advancements in cooling technologies, no review has critically examined the combined effects of two promising passive cooling techniques, radiative paints and metal foam fins, both individually and collectively. This review addresses this gap and offers an integrated approach to these passive cooling methods that has not been sufficiently explored in previous literature. PV modules convert a portion of incident solar light into electricity, while the remainder is converted to heat, increasing module temperature and decreasing power output [6]. The longer the exposure to high temperatures, the greater the likelihood of efficiency loss and material degradation. A rise in temperature by 1 °C results in a 0.45–0.55 % efficiency loss for most panels [7]. Higher temperatures also accelerate material degradation, such as tempered glass degradation at 70–80 °C, which reduces module lifespan [8]. To counteract these impacts, scientists and engineers have explored various thermal regulation methods, ranging from passive devices, such as heat-dissipating fins, to active methods, such as fluid-based cooling and

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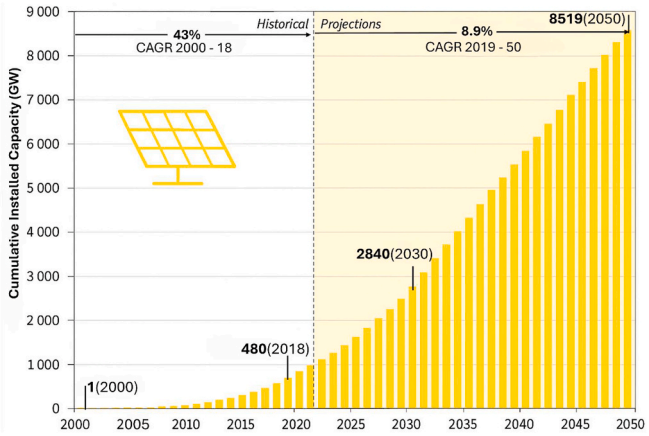


Fig. 1. Yearly projections of solar PV installation capacity from 2000 to 2050 [5].

phase change materials (PCMs) [9]. Cutting-edge solutions, such as integrating nano-materials or coupling PV panels with thermoelectric generators, are emerging as promising avenues for boosting energy output and reliability in diverse environments [10].

Active cooling systems, although improving efficiency, are beset by a succession of challenges. These feature high initial and operational costs, which can render such systems economically unviable, particularly for small installations [11]. The systems also consume additional energy to power pumps and fans, which can negate efficiency gains from lower panel temperatures. [12]. Their installation difficulty is compounded by ongoing maintenance and the need for rugged materials to withstand harsh environmental conditions, further contributing to their operational challenges. Moreover, their dependence on water cooling in low-water-resource environments and their lack of scalability to small-scale systems also hinder large-scale application. [13]. Active cooling systems, though they enhance performance, are constrained by economic, environmental, and operational considerations.

In contrast, passive cooling techniques have emerged as viable alternatives to active methods for managing PV panel temperatures, thanks to their simplicity, cost-effectiveness, and low energy consumption. Among these, radiative paints and metal foam fins show considerable promise. Radiative cooling leverages materials that emit thermal infrared radiation, particularly within the atmospheric transparency window, to dissipate heat. Radiative paints formulated with components like nanoporous films and polymer blends have demonstrated high solar reflectance (up to 94 %) and emittance (over 95 %), enabling cooling powers exceeding 100 W/m^2 and temperature reductions of around 3 °C under certain conditions [14]. These paints offer the advantage of sub-ambient cooling without requiring mechanical systems. On the other hand, metal foam fins, especially those made from open-cell copper or aluminium, enhance convective heat transfer due to their high surface area and permeability. Experimental studies have shown that copper foam fins can reduce PV module temperatures by approximately 8.4 % and boost power output by 4.9 %. In contrast, optimized aluminium fin configurations can achieve temperature drops of up to 5 °C and modest efficiency gains [15,16]. These passive solutions not only improve PV performance but also support long-term reliability without the energy and maintenance burdens associated with active systems.

Despite these improvements, a significant gap remains in combining different cooling techniques to enhance performance. Although many studies have focused on individual cooling methods, no other review has integrated radiative paints with metal foam fins to enable improved cooling performance through synergistic effects. This review proposes such a hybrid approach, offering a comprehensive review of both techniques and their potential when combined for optimal cooling and improved PV efficiency. Table 1 provides an overview of key prior reviews and research on passive cooling of PV panels using radiative paints and metal foam fins, with an emphasis on significant findings. It presents the state of the art of the research. It illustrates that, while each technique has been investigated individually, there is an evident lack of research examining their combined application as a hybrid cooling system.

Despite significant progress in passive cooling technologies for PV panels, critical gaps remain that impede their full potential. Notably, there is a scarcity of studies investigating the synergistic effects of combining radiative paints with metal foam fins, which could unlock enhanced cooling performance through hybrid mechanisms. The optimization of metal foam properties, such as porosity, pore size, thickness, and permeability, under both natural and forced convection conditions remains insufficiently explored, limiting understanding of how these parameters affect convective heat transfer and airflow resistance in PV applications. In addition, experimental inconsistencies and environmental variability have hindered the development of standardized testing protocols, making cross-comparisons challenging. The long-term durability of radiative coatings under UV exposure, humidity, and mechanical stress remains inadequately addressed, raising concerns about their practical longevity. Scalable, cost-effective manufacturing of nanostructured radiative materials also requires further advancement. Moreover, environmental factors such as atmospheric humidity and pollution substantially reduce radiative cooling efficiency but are often overlooked in performance assessments. Field validations at large scales and lifecycle cost analyses for both radiative paints and metal foam fins are also nonexistent, leading to uncertainty about their actual efficiency and economic potential.

Bridging such complex gaps is crucial to developing robust, efficient, and commercially viable passive cooling systems for PV systems. This review addresses these challenges by providing a comprehensive integration of passive cooling methods for PV panels, with particular emphasis on radiative paints and metal foam fins. The key contributions of this work include:

- The review systematically analyses two promising passive cooling methods, such as radiative paints and metal foam fins, highlighting their working principles, performance metrics, and impact on temperature reduction and energy efficiency.
- It provides a detailed assessment of various materials, structural configurations, and performance factors such as porosity, emissivity, and thermal conductivity relevant to PV cooling applications.
- The review introduces an original conceptual framework for combining radiative paints and metal foam fins into a single hybrid system, leveraging both radiative and convective heat transfer mechanisms to enhance PV thermal regulation.
- It identifies key scientific and technical gaps, including durability under environmental stress, lack of standardized testing, and integration constraints. It critically discusses limitations affecting the real-world application of both standalone and hybrid cooling methods.
- The review outlines strategic directions for future work, such as optimizing foam porosity and coating compatibility, performing outdoor validation studies, and addressing economic and material scalability to support practical deployment in diverse climates.

2. Fundamentals of passive cooling in PV panels

2.1. Thermal behavior of PV panels

PV panels convert solar energy into electrical energy; however, a significant portion of the incident solar radiation is transformed into heat, causing an increase in panel temperature and a corresponding decrease in electrical efficiency [25]. This thermal behavior is a critical consideration in the performance and lifespan of PV systems, especially

Table 1
Summary of previous reviews and studies on passive cooling of PV panels using radiative paints and metal foam fins.

Author, Ref.(s)	Study Type	Cooling Enhancers	Focus	Key Results/Findings
Hasan et al. [15]	Experimental	Open-cell copper metal foam fins	Performance enhancement of PV panels	10 longitudinal fins reduced PV panel temperature by 8.4 % and improved power output by 4.9 %
Wong et al. [17]	Experimental	Dual-axis solar tracker, radiative cooling paint, and heat sink	Enhancement of PV performance using passive cooling and tracking systems	Significant improvements in PV efficiency (12.66 % increase) and power output (22.63 % rise) through integration of cooling systems; experimental results confirmed the effectiveness of the cooling enhancements
Lim [18]	Review	Radiative cooling and cool paint	Basic physics, recent developments, and potential impacts	Discusses the effectiveness of radiative cooling and cool paint in reducing building heat gain and air-conditioning usage
Grubišić-Čabo et al. [19]	Experimental	Aluminum fins	Energy conversion efficiency improvement	Randomly positioned perforated L-profile fins improved efficiency by 2 % and enhanced the cooling rate.
Carne et al. [20]	Experimental	Radiative cooling paint (colored: green and blue)	Enhanced solar reflectance and UV resistance	Colored paints showed up to 6.3 °C lower temperatures than commercial paints; they also improved UV resistance due to inorganic pigments.
Sato et al. [21]	Review	Various PV cooling methods, including radiative cooling	Overview of PV cooling technologies; quantitative evaluation of radiative cooling	Radiative cooling's enhancement is limited by surface emissivity modification under typical climates; it guides future research
Perrakis et al. [22]	Modeling & Review	Radiative cooling coatings, photonic structures	Comparative study of radiative cooling and photonic approaches for PV temperature control	Combining radiative cooling with sub-bandgap reflection enhances efficiency; guidelines for photonic cooler design for PV panels.
Variji et al. [23]	Numerical Study	Metal foam in PCM	Impact of metal foam porosity on PV-PCM systems	Metal foam with 0.9 porosity enhanced temperature reduction and electrical efficiency by 6.8 % and 9.8 %.
Murtadha et al. [24]	Experimental	Aluminium fins (various arrangements)	Effects of fin arrangement and forced air cooling on PV performance	Longitudinal fins with forced air-cooling reduced surface temperature by 11.86 % and increased output power by 5.42 % compared to baseline
Present study	Review	Radiative paints and metal foam fins	Comprehensive critical review of passive cooling methods and materials for PV panels	Provides an integrated synthesis of thermal mechanisms, materials, configurations, and performance; identifies scientific gaps; proposes future research directions.

in regions with high solar irradiance or limited natural cooling. Understanding the underlying thermal dynamics helps guide the development of cooling strategies aimed at maintaining optimal operating conditions [26]. This interaction between solar input, thermal dissipation, and electrical output is illustrated in Fig. 2, which shows the interconnected thermal, radiation, and electrical models used to evaluate PV performance.

When solar radiation strikes a PV module, only a small fraction, typically 15–20 % is converted into electricity, while the majority is absorbed as thermal energy. This heat accumulation arises from the absorption of infrared radiation and the recombination losses within the PV cells. [28]. Under intense sunlight, cell temperatures can exceed 50 °C, significantly lowering the panel's output [29]. Several external and internal factors affect the thermal response of PV panels, including ambient temperature, solar irradiance, wind speed, humidity, and mounting configuration. High ambient temperatures and strong irradiance levels raise cell temperature, whereas wind facilitates convective cooling [30]. Dust accumulation and moisture can further exacerbate heating by modifying the surface thermal properties and reducing efficiency [31].

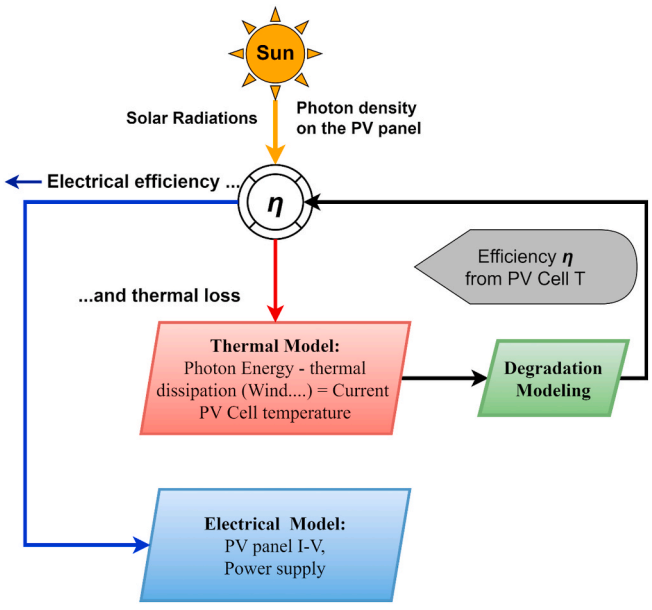


Fig. 2. Schematic representation of the energy conversion process in PV panels, showing how solar radiation is partially converted into electrical energy while excess energy is dissipated as heat [27]. (Reproduced with permission from the publisher).

The most direct consequence of elevated temperature is the reduction in electrical efficiency. For monocrystalline silicon PV cells, efficiency drops by approximately 0.4 % for every 1 °C increase in temperature [29]. This relationship is commonly represented by the temperature-corrected efficiency equation [32]:

$$\eta_T = \eta_{ref}\left[1 - \beta\left(T_{cell} - T_{ref}\right)\right] \quad (1)$$

where η_T is the panel efficiency at operating temperature T_{cell}, η_{ref} is the reference efficiency at 25 °C, and β is the temperature coefficient (typically 0.003–0.005/°C). As such, a 25 °C rise in cell temperature could reduce panel efficiency by 10 % or more. Fig. 3 visually demonstrates how the electrical efficiency of PV panels decreases with increasing temperature, reinforcing the need for effective thermal regulation strategies. This temperature-induced degradation is not only a performance issue but also poses long-term risks due to thermal fatigue, material delamination, and solder joint failure [27].

Heat transfer within PV panels occurs via three primary mechanisms: conduction, convection, and radiation. Conduction transfers heat between layers of the panel, such as from the glass to the backsheet [33]. Convection, including both natural and forced airflow, is the dominant heat-loss mechanism in many outdoor installations [34]. Radiation also plays a significant role, especially during night cooling or in dry climates, where the panel surface emits infrared energy into the sky. Effective cooling solutions must account for these mechanisms to enhance heat dissipation without significant additional energy consumption.

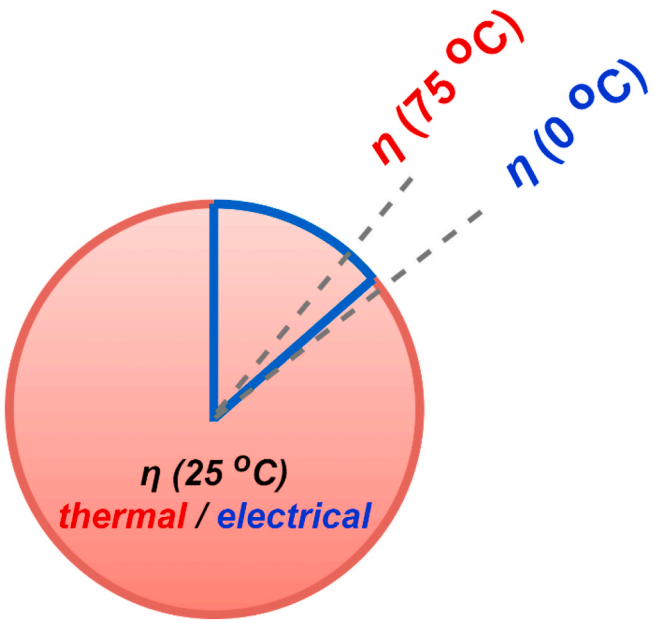


Fig. 3. Graphical depiction of PV panel efficiency at different temperatures. The efficiency drops from its maximum at 25 °C as cell temperature rises to 75 °C, and increases slightly when operating below standard test conditions, such as at 0 °C. Adapted from Ref. [27]. (Reproduced with permission from the publisher).

2.2. Passive cooling mechanisms

2.2.1. Radiative cooling: Principles and thermal balance

Radiative cooling is a passive thermal management technique that enables surfaces to reject heat directly to the cold outer space, which has an effective temperature of approximately 3 °C. This phenomenon is facilitated by the emission of infrared radiation, particularly within the atmospheric transparency window of 8–13 μm, through which the Earth's atmosphere allows thermal radiation to escape without significant absorption [35]. As a result, materials engineered for radiative cooling can maintain surface temperatures below ambient, even under direct solar illumination, without consuming external energy [36].

The theoretical foundation of radiative cooling is governed by Kirchhoff's law of thermal radiation, which states that any object at a temperature above absolute zero emits and absorbs electromagnetic radiation [37]. For effective net cooling, the surface must exhibit high thermal emissivity within the mid-infrared range (8–13 μm) and simultaneously possess high solar reflectance in the 0.3–2.5 μm wavelength range to minimize solar heat gain [38]. This dual optical requirement enables the surface to efficiently emit thermal radiation while avoiding unwanted heating from solar radiation.

The overall thermal balance of a radiative cooling surface (RCS) can be described using an energy equation that accounts for radiative, convective, and conductive heat exchanges [38]. The net cooling power per unit area, Q_{net}, is given as [36]:

$$Q_{net} = Q_{rad}(T_{surf}) - Q_{sol} - Q_{atm}(T_{amb}) - Q_{conv} - Q_{cond} \quad (2)$$

where:

Q_{rad} is the surface emitting the thermal radiation,

Q_{sol} is the absorbed solar radiation,

Q_{atm} does the surface absorb the atmospheric longwave radiation,

Q_{conv} and Q_{cond} represent the heat exchanged by convection and conduction, respectively

T_{surf} is the temperature of the radiative cooling surface,

T_{amb} is the temperature of the surrounding ambient environment (air temperature).

The emitted radiative power Q_{rad} is further defined as [36]:

$$Q_{rad}(T_{surf}) = A \int \cos\theta \, d\Omega \int_0^{\infty} I_{bb}(\lambda, T_{surf}) \, \varepsilon_{surf}(\lambda, \Omega) \, d\lambda \quad (3)$$

$$I_{bb}(\lambda, T) = \frac{2hc^2}{\lambda^5} \left[\frac{1}{exp\left(\frac{hc}{\lambda kT}\right) - 1} \right] \quad (4)$$

In these expressions, I_{bb} represents the spectral radiance of a blackbody at wavelength λ and temperature T, ε_{surf} is the spectral and directional emissivity of the surface, and A is the radiating area and θ, Ω are the zenith and solid angles, respectively.

The absorbed atmospheric longwave radiation is expressed as [36]:

$$Q_{atm} = A \int \cos\theta d\Omega \int_0^{\infty} \varepsilon_{surf}(\lambda, \Omega) \, \varepsilon_{atm}(\lambda, \Omega) \, I_{bb}(\lambda, T_{atm}) \, d\lambda \quad (5)$$

The atmospheric spectral emissivity, $\varepsilon_{atm}(\lambda, \Omega)$, on the transmittance $t(\lambda)$ and is given by Ref. [39]:

$$\varepsilon_{atm}(\lambda, \Omega) = 1 - t(\lambda)^{1/\cos\theta} \quad (6)$$

The solar radiation absorbed by the cooling surface is determined from Ref. [36]:

$$Q_{sol} = A \int_{0.3}^{2.5} \varepsilon(\lambda, \theta_{sol}) \, I_{sol}(\lambda) \, d\lambda \quad (7)$$

where $\varepsilon(\lambda, \theta_{sol})$ is the angular-dependent spectral absorptance of the surface and $I_{sol}(\lambda)$ is the spectral solar irradiance.

Convective heat loss between the surface and ambient air is described by Newton's law of cooling [40]:

$$Q_{conv} = hA(T_{amb} - T_{surf}) \quad (8)$$

where h is the convective heat transfer coefficient $(W \cdot m^{-2} \cdot K^{-1})$, influenced by wind speed and meteorological conditions.

Conductive heat transfer between the cooling surface and adjacent materials is calculated as [40]:

$$Q_{cond} = kA(T_{obj} - T_{surf}) \quad (9)$$

where k is the thermal conductivity of the contacting material, and T_{obj} is its temperature.

These equations, together, constitute a complete thermophysical model for evaluating the performance of radiative cooling surfaces under different environmental and design conditions. Mohamadkhani [36] further emphasizes the significance of the atmospheric window for achieving maximum radiative heat rejection. As seen in Fig. 4(a), the 8–13 μm wavelength range is strongly resistant to atmospheric absorption, allowing the emitted radiation to pass freely into space. Fig. 4(b) is a conceptual illustration of heat transfer processes impinging on an RCS, such as solar irradiation, atmospheric back-radiation, and non-radiative conduction and convection losses. The illustrations highlight the importance of spectral selectivity and surface isolation in optimizing passive radiative cooling.

2.2.2. Radiative paints and selective emitters for PV cooling

Passive radiative cooling (PRC) has gained substantial attention as a sustainable, energy-free method to reduce PV module operating temperatures by enhancing thermal emission through the atmospheric transparency window (8–13 μm) [41,42]. By decreasing the temperature of PV panels, PRC improves power conversion efficiency (PCE) and module longevity. This section reviews the principles, materials, and applications of selective emitters and radiative coatings for PV cooling, alongside comparative performance, environmental considerations, and practical challenges.

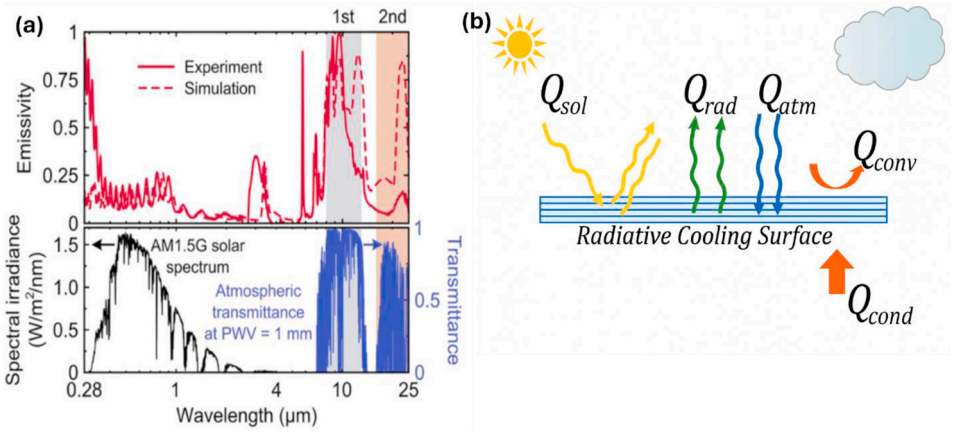


Fig. 4. (a) Spectral atmospheric transmittance showing the 8–13 μm window for radiative heat rejection. (b) Schematic of heat transfer mechanisms in a radiative cooling surface (RCS) [36]. (Reproduced with permission from the publisher).

2.2.2.1. Selective emitters and photonic structures. Selective emitters are engineered materials or nanostructures that maximize thermal emissivity within the atmospheric window while minimizing absorption of solar radiation, thereby maintaining high PV efficiency [43]. Photonic structures such as multilayer thin films, photonic crystals, and metamaterials manipulate light-matter interactions to achieve the desired spectral selectivity. The fundamental heat transfer mechanisms involved in PV module thermal management are illustrated in Fig. 5. This schematic highlights the principal thermal processes, such as solar absorption, thermal emission, conduction, and convection, and the two main strategies to reduce PV temperature: limiting parasitic solar absorption and enhancing radiative emission through the atmospheric window [44].

Multilayer dielectric films composed of alternating materials with contrasting refractive indices (such as SiO_2, TiO_2, Al_2O_3) have demonstrated temperature reductions of up to approximately 6 °C by improving both solar reflectance and infrared emissivity [45]. Photonic crystals with periodic nanoscale features create photonic band gaps that selectively enhance emission within the atmospheric window, resulting in temperature reductions of 12–18 °C. A compelling example is the silica pyramid photonic structure, which combines high solar reflectance and selective thermal emission for enhanced radiative cooling [43, 46]. As illustrated in Fig. 6(a), the schematic compares layered structures for bare silicon, an ideal thermal emitter with perfect emissivity, a 5 mm silica film, and the silica pyramid photonic structure. The emissivity and absorptivity spectra shown in Fig. 6(b) highlight the superior emissivity of the silica-based emitters, particularly within the 8–13 μm atmospheric window. Temperature profiles in Fig. 6(c)–(e) demonstrate the cooling performance of these structures under varying solar heating power and convective heat transfer coefficients, with the silica pyramid structure consistently achieving the lowest operating temperatures. This validates the efficacy of photonic designs in PRC for PV applications.

Complementing the performance of silica-based photonic structures, Li et al. [48] provide a comprehensive overview of advanced photonic designs for passive radiative cooling, as illustrated in Fig. 7. These include multilayer dielectric stacks, photonic metamaterials, and hierarchically structured surfaces tailored to enhance emission within the atmospheric transparency window while minimizing solar absorption. Fig. 7(a) presents the spectral emissivity profile of a nanostructured photonic crystal, optimized for high emissivity in the mid-infrared region and minimal absorption in the solar spectrum. Cross-sectional views in Fig. 7(b) and the schematic in Fig. 7(c) illustrate multilayer dielectric stacks composed of HfO_2, SiO_2, Al_2O_3, and Ag, engineered for spectrally selective emission. The photonic metamaterial array shown in Fig. 7(d) represents a geometric arrangement of Ge-Al resonators that facilitates spectral tuning.

2.2.2.2. Radiative paints and coatings. Radiative paints and coatings provide a practical, scalable, and low-cost strategy to passively cool PV panels by enhancing mid-infrared (mid-IR) emissivity and solar reflectance [49]. These coatings typically incorporate polymers, inorganic nanoparticles, or composite thin films that are spectrally selective, minimizing solar absorption (0.3–2.5 μm) while maximizing thermal emission within the atmospheric transparency window (8–13 μm) [50].

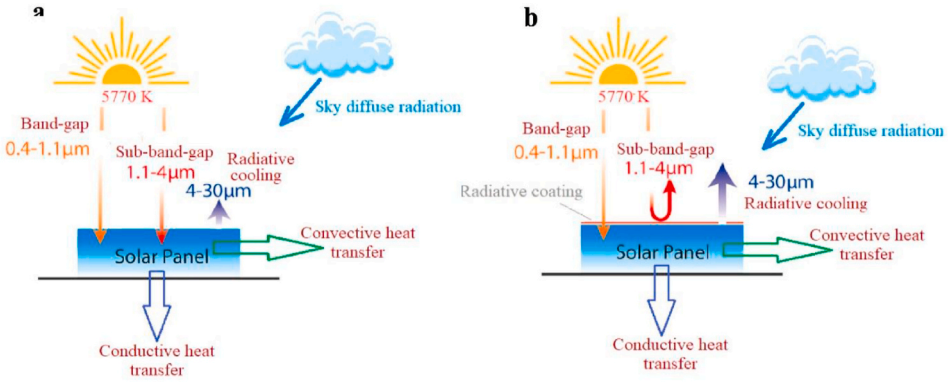


Fig. 5. Schematic representation of heat transfer mechanisms in a PV module without (a) and with (b) a radiative cooling coating. The incident solar spectrum is divided into band-gap (0.4–1.1 μm) and sub-band-gap (1.1–4 μm) radiation. The radiative cooling coating enhances thermal emission in the atmospheric transparency window (4–30 μm), promoting passive cooling. Other heat transfer modes include convective and conductive heat transfer, as well as sky diffuse radiation [44]. (Reproduced with permission from the publisher).

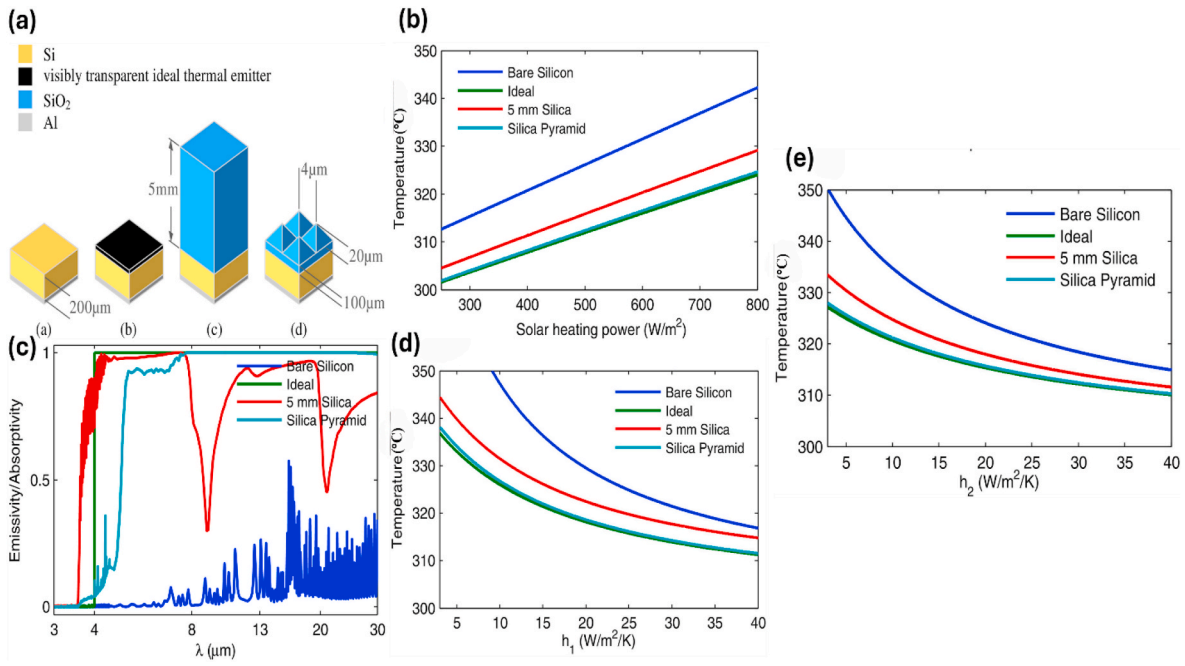


Fig. 6. Design and thermal performance evaluation of various selective thermal emitter structures for PV cooling. (a) Schematic cross-sectional illustrations of four emitter configurations: bare silicon, an ideal thermal emitter with perfect emissivity, a five mm-thick silica film, and a silica pyramid photonic structure engineered for spectral selectivity; (b) Corresponding spectral emissivity and absorptivity curves for each configuration, highlighting the enhanced emission of silica-based structures within the atmospheric transparency window (8–13 μm); (c) Predicted steady-state temperatures of the emitters as a function of solar heating power, demonstrating the superior cooling capability of the silica pyramid design; (d) and (e) Temperature response as a function of convective heat transfer coefficients h_1 and h_2, respectively, under different environmental conditions [44,47]. (Reproduced with permission from the publisher).

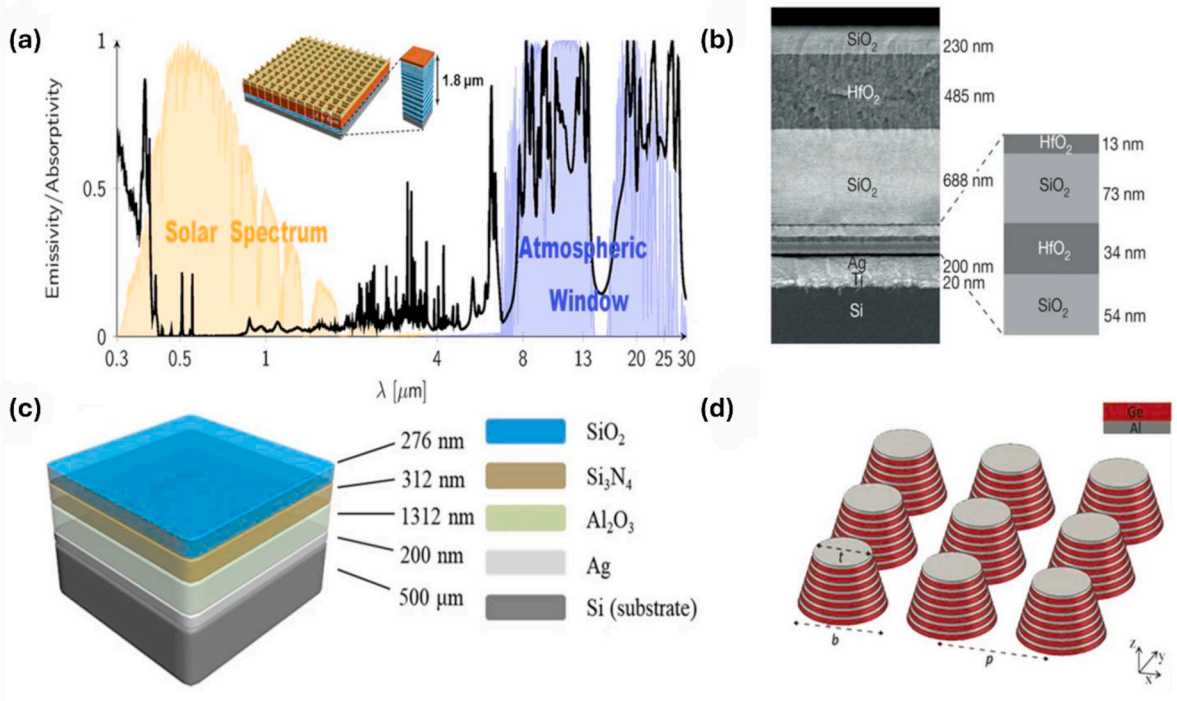


Fig. 7. Advanced photonic and hierarchical designs for passive radiative cooling. (a) Spectral emissivity/absorptivity profile of a nanostructured photonic emitter designed for high mid-IR emission and minimal solar absorption, with inset showing a schematic of the structure; (b) Cross-sectional SEM image of a multilayer structure composed of HfO_2 and SiO_2 films; (c) Schematic of a dielectric multilayer stack including SiO_2, Si_3N_4, Al_2O_3, Ag, and a Si substrate; (d) 3D schematic of a Ge-Al photonic metamaterial array [48]. (Reproduced with permission from the publisher).

Polymers such as polydimethylsiloxane (PDMS) and polymethylpentene (TPX) are commonly used due to their high thermal emissivity (typically 0.9–0.95) and optical transparency across the visible and near-infrared spectrum [51]. As illustrated in Fig. 8, microstructured coatings comprising triangular PDMS arrays deposited on a SiO2/Ag substrate exhibit enhanced radiative performance. The schematic in Fig. 8(a)

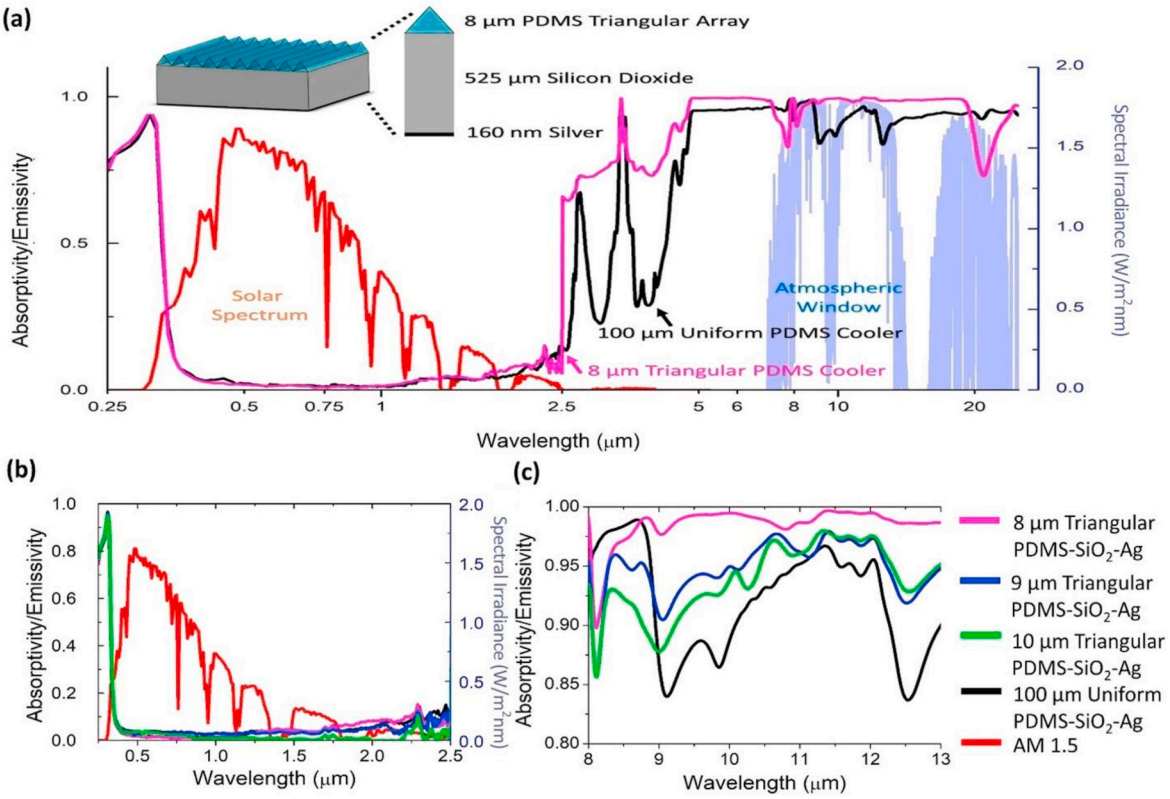


Fig. 8. Microstructured polymer-based radiative coatings for passive cooling applications. (a) Schematic cross-section of an 8 µm triangular PDMS array deposited on a silicon dioxide (SiO_2) layer atop a silver (Ag) substrate, designed to enhance mid-infrared emissivity while minimizing solar absorption; (b) Absorptivity and emissivity spectra of various PDMS structures over the solar spectrum (0.25–2.5 µm), demonstrating low solar absorptivity essential for effective cooling; (c) Emissivity spectra within the atmospheric transparency window (8–13 µm) for different PDMS geometries and thicknesses, highlighting superior thermal emission from the triangular microstructured array compared to uniform coatings [52]. (Reproduced with permission from the publisher).

shows an 8 µm PDMS triangular surface structure backed by silver and a thick SiO_2 layer. Spectral analysis confirms that the triangular array maintains low absorptivity across the solar spectrum while achieving high emissivity in the mid-IR, particularly within the 8–13 µm atmospheric window. Fig. 8(b) and (c) compare the spectral response of various PDMS structures, revealing that triangular patterns significantly outperform uniform coatings in both mid-IR emissivity and solar rejection [52]. These engineered microtextures help maximize radiative heat loss while minimizing solar gain, thereby improving overall cooling effectiveness.

In addition to polymer-based coatings, pigmented formulations incorporating inorganic nanoparticles such as titanium dioxide (TiO_2) and barium sulphate ($BaSO_4$) are widely utilized to enhance solar reflectance [53]. These pigments act as highly efficient scatterers with minimal absorption across the visible to near-infrared spectrum, effectively reducing solar heat gain and achieving temperature reductions of over 14 °C in some cases [54]. Further enhancements in passive cooling performance arise from the incorporation of nanoparticles directly into polymer matrices, forming composite coatings that synergistically combine strong scattering with high mid-infrared emissivity. For instance, SiO_2 nanospheres embedded within TPX matrices, as demonstrated in Fig. 9, exhibit an infrared emissivity near 0.85 while maintaining excellent visible light transmittance above 90 % [54]. This unique balance of thermal emissivity and optical transparency makes these nanocomposite coatings highly suitable for PV encapsulation and surface applications, enabling improved thermal regulation without compromising PV efficiency or light-harvesting capability.

2.2.2.3. Comparative performance and environmental considerations. PRC coatings for PV panels exhibit a wide range of performance, depending on material composition, structural design, environmental conditions, and integration approaches. As summarized in Table 2, these coatings vary significantly in their optical properties, achievable temperature reductions, and impacts on PCE. The following discussion highlights key comparative performance metrics and environmental factors influencing PRC effectiveness.

Simple polymer composite coatings, such as those incorporating SiO_2 microparticles embedded in a polymer matrix, provide modest temperature reductions of approximately 1–1.2 °C and corresponding efficiency gains near 0.21 % [55]. These coatings typically achieve solar reflectance of 85–90 % and mid-infrared emissivity of 0.90–0.92. Their predominant rear-surface application on double-glass PV panels minimizes optical losses and maintains module transparency. More sophisticated hybrid systems integrate PV, photothermic, and radiative cooling functions. For instance, polyethylene films enable mid-infrared transmittance of roughly 85–95 % and facilitate significant radiative cooling power (~72 W/m^2) under outdoor conditions, although explicit steady-state temperature reductions are often not detailed due to the complex interaction of convective and radiative heat transfer processes [56]. Advances in micro- and nanostructured coatings, including transparent radiative heat mirrors (TRHM) composed of ITO layers and honeycomb PDMS microstructures, achieve substantial temperature reductions (~7.2 °C) and efficiency improvements (~0.81 %) while maintaining high optical transparency and solar rejection (reflectance ~65 % in the near-infrared, emissivity ~93 % in the mid-infrared) [58].

Multilayer antireflective coatings (ARCs) that combine Si_3N_4, SiO_2, and PDMS effectively balance solar absorption and thermal emission, reducing module temperatures by about 11.6 °C and improving efficiency by roughly 0.56 %. Meanwhile, highly porous double-layer coatings featuring BK7 glass show some of the most noticeable cooling

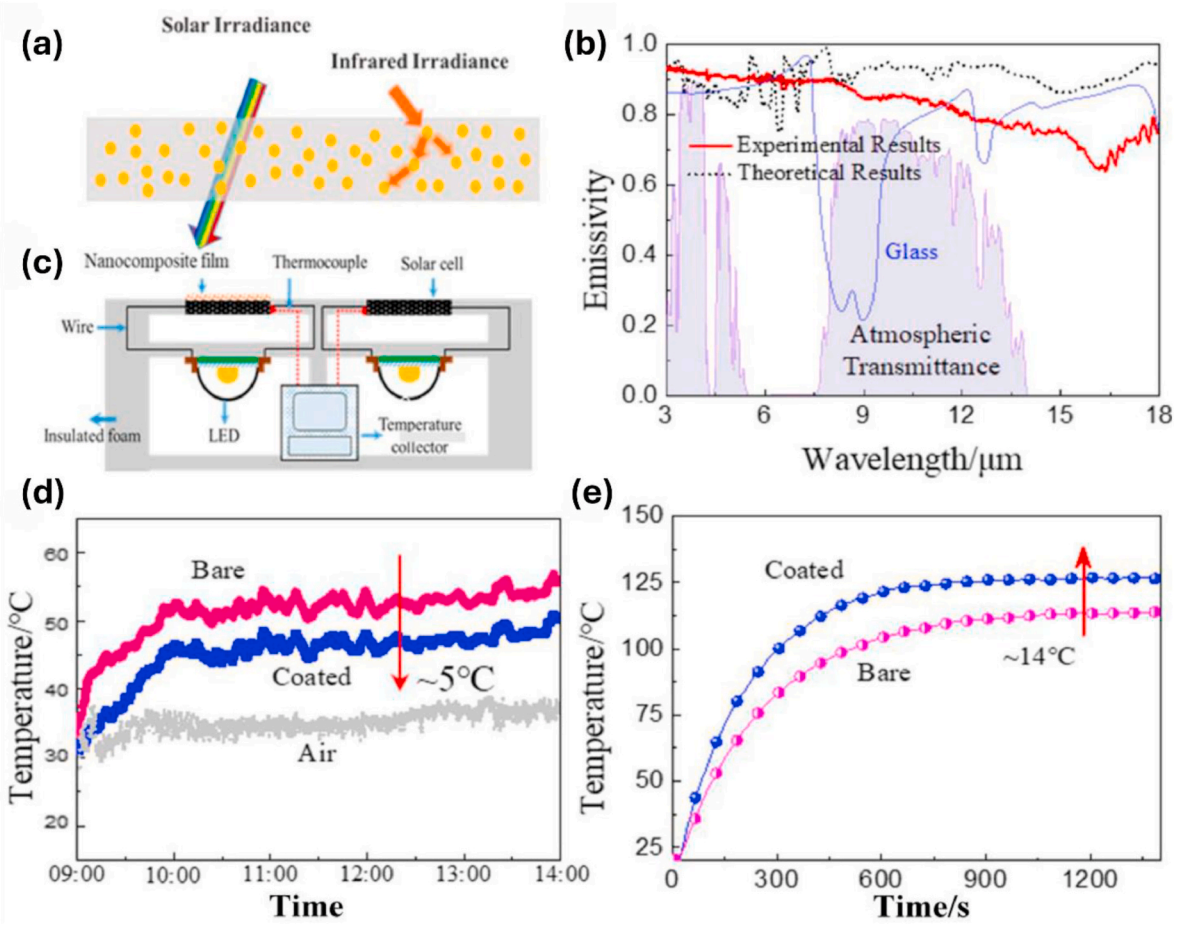


Fig. 9. Characterization and outdoor performance of SiO_2 nanosphere–TPX nanocomposite coatings for PV cooling. (a) Schematic illustrating solar and infrared irradiance interaction with the nanocomposite film; (b) Measured and theoretical spectral emissivity of the coating, highlighting strong emission within the atmospheric transmittance window; (c) Experimental setup including coated and bare solar cells equipped with thermocouples and LED loads for temperature monitoring; (d) Temperature profiles over time demonstrating approximately 5 °C reduction in surface temperature of the coated solar cell compared to the bare counterpart under outdoor conditions; (e) Real-time temperature measurement of LEDs powered by coated and bare solar cells, showing a ~14 °C higher equilibrium temperature for the LED powered by the coated cell, indicating enhanced power output due to improved thermal management [54]. (Reproduced with permission from the publisher).

effects, with temperature drops exceeding 18 °C and efficiency gains above 20 %, mainly under controlled laboratory or optimized installation conditions. Nanoporous AAO films, characterized by around 65 % porosity and nanoscale pores (~94 nm), deliver outstanding radiative cooling performance. They achieve temperature reductions of about 7.5 °C in flat-panel silicon cells and up to roughly 35.6 °C under concentrated sunlight, resulting in efficiency improvements of up to 16 %. Similarly, $Y2O_3/TiO_2$ microparticle-embedded PDMS coatings exhibit high solar reflectance (~92 %) and mid-infrared emissivity (~95 %), producing temperature reductions near 7.7 °C in outdoor PV applications, highlighting their potential for integrated energy-saving solutions.

Environmental and climatic factors greatly influence PRC effectiveness. High atmospheric humidity, cloud cover, and aerosol levels decrease the transmittance of the atmospheric window (8–13 μm), thus limiting radiative heat loss potential [49]. Cooling performance is usually lower on humid or overcast days compared to clear, dry days. Additionally, convective heat transfer, driven by wind speed, affects module temperatures. Increased airflow improves convective cooling and can also reduce net radiative cooling by raising the convective heat transfer coefficient. Therefore, installation designs that optimize ventilation and airflow around PV panels are vital for maximizing thermal management efficiency [63].

Durability and environmental stability, including UV resistance, mechanical robustness, and contamination resilience, are critical for long-term coating performance. Multifunctional coatings incorporating self-cleaning properties, such as those containing ZnO, SiO_2, and chlorophyll, maintain optical clarity and thermal emissivity by mitigating dust accumulation, which otherwise degrades cooling effectiveness and module output [63]. In conclusion, radiative cooling coatings demonstrate a broad spectrum of performance, influenced by materials engineering and environmental conditions. Continued advancements in spectral design, environmental adaptability, and durability will be vital for widespread adoption, contributing significantly to enhanced PV efficiency and module longevity.

2.2.2.4. Limitations and challenges. PRC coatings and selective emitters for PV cooling face several limitations that must be addressed to enable broader adoption and sustained performance. One major challenge arises from environmental factors such as ambient temperature, wind speed, solar intensity, humidity, and cloud cover, which all affect cooling efficiency. For instance, high atmospheric water vapor and cloud cover reduce the transparency of the atmospheric window (8–13 μm), thereby limiting radiative heat rejection and reducing cooling effectiveness, especially in humid or overcast conditions [64,65]. Additionally, convective heat transfer can interfere with net radiative cooling; while increased airflow can enhance convective heat loss, it may simultaneously raise the convective heat transfer coefficient and

Table 2
Comparative overview of radiative cooling coatings for PV panels.

Material Type	Composition/Structure	Solar Reflectance (R_sol)	Mid-IR Emissivity (ε_8-13 μm)/IR Transmittance	Temperature Reduction (ΔT, °C)	Efficiency Improvement (%)	Application Location	Reference (s)
Spectral regulation coating	SiO_2 microparticles (~1 μm) in polymer matrix	~0.85–0.90	~0.90–0.92	~1.0–1.2	~0.21	Rear surface	[55]
Hybrid PV-photothermic-radiative cooling system	Al substrate, black TPT, Si solar cells, transparent TPT, polyethylene film (6 μm)	_	Polyethylene IR transmittance ~0.85–0.95	Cooling power ~72 W/m^2 at night	_	Rooftop integrated	[56]
Nanoporous anodic aluminium oxide (AAO)	Nanoporous AAO film (porosity ~0.65), hole diameter ~94 nm, fabricated by two-step anodization	Solar transmittance ~0.94 (implying solar reflectance ~0.06 or less).	~0.97	~7.5 (flat panel), ~35.6 (concentrated PV)	~2.72 (flat panel), ~16 (concentrated PV)	Solar cell encapsulation	[57]
Transparent-radiative heat mirror (TRHM)	ITO layer + honeycomb-like PDMS microstructure coating	~0.65 (NIR 1.1–2.5 μm)	~0.93	~7.2	0.81	PV module surface	[58]
Multilayer ARC coating	Si_3N_4/SiO_2/PDMS multilayer coating	_	Radiative emission ~346 W/m^2 at 35 °C ambient	~11.6	~0.56	Crystalline PV module surface	[59]
Double-layer porous/BK7 coating	Porous top layer + BK7 glass bottom layer	High absorption (0.3–1.1 μm)	>0.96	~18.04	21.56	Monocrystalline Si solar cell surface	[60]
Multifunctional nanocomposite coating	ZnO, SiO_2, and chlorophyll-based coating	Visible transmittance ~93–96 %	Optimized (highest emissivity)	~9.62	22.12	Solar panel surface	[61]
Y_2O_3/TiO_2 microparticle-embedded PDMS coating	Y_2O_3 and TiO_2 microparticles embedded in PDMS polymer matrix	150	Y_2O_3: ~4–8 μm; TiO_2: ~0.3 μm	~7.7	_	Energy-saving PV surfaces	[62]

reduce sub-ambient temperature drops [66].

Material and fabrication challenges also hinder the large-scale deployment of photonic selective emitters. The complex nano-fabrication techniques required for precise spectral selectivity, such as anodization and DC-sputtering deposition, limit scalability and increase costs [67]. Furthermore, polymer-based radiative coatings, commonly used for their high mid-infrared emissivity and optical transparency, exhibit durability issues. These polymers are susceptible to UV degradation, mechanical wear, and environmental exposure, including high temperature, moisture, and abrasion, leading to gradual declines in cooling performance and material integrity [68].

Real-world field studies demonstrate that long-term exposure to weathering can cause yellowing, cracking, and peeling of radiative coatings, further reducing their efficiency over time [69]. Tropical climates with intense solar radiation and humidity present particularly harsh conditions that challenge sustained sub-ambient cooling [70]. Maintenance and cleaning are thus critical to mitigate performance degradation caused by dust accumulation and contamination, which can significantly impair radiative heat loss [71]. Integration of radiative cooling coatings with PV panels introduces additional complexities. While reduced module temperatures can enhance PCE and extend panel lifespan, the coatings must not compromise optical transparency or electrical characteristics. Material compatibility, thermal cycling effects, and potential hotspots require careful design considerations [72].

From an economic perspective, the cost-effectiveness of radiative cooling coatings depends on the materials used, the scalability of fabrication, and long-term durability. Polymers offer lower upfront costs but suffer from limited lifespan, while ceramic or inorganic nanoparticle-based coatings provide superior durability at a higher cost [73]. Innovations in waterborne and environmentally friendly paints show promise for scalable, cost-efficient applications, but balancing performance, durability, and cost remains an ongoing challenge [74]. Despite promising laboratory results, the widespread adoption of PRC coatings for PV systems is limited by environmental variability, fabrication complexity, durability issues, integration challenges, and cost considerations. Addressing these challenges through advanced materials research, scalable fabrication methods, and robust outdoor testing is crucial for realizing the full potential of this sustainable cooling technology.

2.2.3. *Metal foam fins for enhanced convective cooling*

PV panels experience efficiency losses and accelerated degradation at elevated operating temperatures. Passive cooling through metal foam fins offers a promising solution by leveraging their porous open-cell architecture to enhance convective heat transfer with the surrounding air [75]. The high surface area, combined with increased airflow penetration, improves natural convection, thus reducing panel temperature without added energy input or significant weight increase [76]. The intrinsic high thermal conductivity of aluminium and copper metal foams further facilitates heat spreading, making them suitable for PV thermal management [77].

2.2.3.1. *Classification and structural characteristics of metal foam fins.*

Metal foam fins have attracted significant interest in thermal management systems because of their unique porous morphology, which enhances heat transfer and fluid flow. Metal foam materials are generally categorized according to their pore structure, porosity, and material, each of which is a deciding factor for their thermal and mechanical performance [78]. Open-cell metal foams with interconnected pores enhance convective heat transfer by enabling fluid flow and maximizing heat exchange surface area [79]. In contrast, closed-cell foams possess isolated pores that improve mechanical strength and thermal insulation but limit fluid permeability and convective efficiency [80].

Among the materials used for metal foam fins, aluminium foams are favored for their lightweight and relatively high thermal conductivity, making them suitable for applications requiring both heat dissipation and structural integration, such as in PV panel cooling [81]. Copper foams exhibit superior thermal conductivity and electrical properties, which can offer enhanced heat transfer performance in PV cooling systems where maximizing thermal conductivity is critical [82]. The choice between aluminium and copper foams often depends on balancing thermal performance with weight constraints and cost considerations.

Although existing studies predominantly evaluate metal foam fins

for heat exchangers and electronic cooling, the fundamental properties of these foams, such as high porosity (typically 70–90 %), pore density (measured in pores per inch, PPI), and material thermal conductivity, provide a promising basis for their use in PV passive cooling applications. High PPI values increase the surface area and turbulence within the foam, promoting efficient heat dissipation but also increasing airflow resistance; thus, design optimization is essential to achieve maximum cooling effect without excessive pressure drop [83]. While the literature lacks direct studies of metal foam fins in PV cooling, the classification and structural characteristics of aluminium and copper metal foams position them as excellent candidates for enhancing passive cooling. Leveraging these materials can effectively reduce PV operating temperatures, improve electrical efficiency, and extend module lifespan. Table 3 highlights the diverse range of metal foam fins studied in the literature, emphasizing their porosity, pore size, thermal conductivity, density, permeability, and typical applications relevant to PV passive cooling. Analyzing these parameters provides valuable insights into selecting and optimizing metal foam fins for PV thermal management.

Open-cell aluminium foams dominate as preferred materials due to their favorable balance of high porosity (ranging from approximately 68 %–97 %), moderate density (150–300 kg/m^3), and thermal conductivity generally between 25 and 60 W/m·K [76,87,89]. This combination offers an optimal trade-off between effective heat dissipation and lightweight design, critical for integrating fins without adding significant structural load to PV panels. Porosity levels above 85 % are common and contribute to large surface areas that enhance convective cooling, although the associated increase in airflow resistance demands careful consideration in system design to avoid excessive pressure drops [86,90].

Copper foams, though less prevalent in PV-specific studies, exhibit considerably higher thermal conductivity (~200–210 W/m·K) and moderate densities (300–600 kg/m^3), as observed in electroformed and open-cell variants [82,84]. Their superior thermal performance makes them excellent candidates for high-demand cooling applications where maximizing heat removal is paramount. However, their higher density and potential cost implications may limit widespread adoption compared to aluminium foams, especially in large-scale PV systems where weight and cost are critical constraints.

Other metal foams, such as nickel and hybrid aluminium-copper foams, display varying porosity (85–95 %) and permeability ranges that influence their convective heat transfer capabilities and mechanical properties [85,88]. The reported permeability values span several orders of magnitude, reflecting differences in pore size and foam structure that directly affect airflow resistance and, in turn, cooling efficiency.

It is noteworthy that foams with finer pores (higher PPI values) generally offer increased surface area and heat transfer potential, but at the cost of reduced permeability and increased pressure drop, underscoring the importance of optimizing pore size relative to the airflow regime and cooling requirements [83,90]. For PV passive cooling, where airflow is primarily natural or low-velocity forced convection, selecting metal foams with a balanced pore size (10–20 PPI) appears advantageous to maximize heat transfer while minimizing fan power or aerodynamic losses.

However, aluminium-based open-cell metal foams with porosity of 85–95 %, pore sizes corresponding to 10–20 PPI, and moderate thermal conductivity (~30–50 W/m·K) are the most practical and effective materials for PV passive cooling applications. Copper foams provide superior thermal conductivity but must be weighed against their higher density and cost. Future PV cooling system designs should consider these material properties alongside aerodynamic and structural factors to optimize thermal management and improve overall PV module performance and durability.

2.2.3.2. Heat transfer mechanisms in metal foam fins. Metal foam fins exhibit complex heat transfer behavior due to their unique porous structures, in which conduction, convection, and radiation interact to influence overall thermal performance. Conduction within metal foams occurs through both the solid metal ligaments and the fluid occupying the pores. The effective thermal conductivity (ETC) of metal foams is highly dependent on key parameters, including porosity, pore size, and temperature. Generally, an increase in porosity and reduction in pore size tend to lower the ETC because the solid phase available for conduction decreases, while the tortuous path enhances thermal resistance within the foam [91]. However, due to the interconnected structure, the ETC of metal foams can exceed expectations by facilitating multi-path conduction, especially when considering the combined solid-fluid matrix [92].

Convection is significantly enhanced in metal foam fins due to the large internal surface area and complex pore geometry, which promotes vigorous fluid mixing and disruption of thermal boundary layers. This results in increased convective heat transfer coefficients compared to traditional solid fins [78]. Forced convection inside the foam structure leads to better fluid–solid interaction, increasing heat dissipation efficiency in applications such as heat exchangers and passive cooling devices [93].

At elevated temperatures, radiative heat transfer within metal foams

Table 3
Comprehensive structural and material properties of metal foam fins relevant to PV passive cooling.

Metal Foam Type	Porosity (%)	Pore Size (mm)/ PPI	Thermal Conductivity (W/m·K)	Density (kg/ m^3)	Permeability (m^2)	Typical Use	Reference (s)
Open-Cell Aluminum Foam	68.2	~10 PPI (~2.54 mm)	~50	150–300	1.67×10^{-7} to 5.91×10^{-7}	Water-cooled PV heat sink	[76]
Copper foam (electroformed)	59.5–86.8	0.635–2.54 (10–40 PPI)	~200	~300–600	$0.6–2 \times 10^{-9}$	High-performance cooling	[84]
Aluminum Foam (Al 85-20)	85	0.20	–	–	1.2×10^{-8} (air)	Common aluminium foam	[85]
Nickel Foam (Ni 100-45)	95	0.45	–	–	5.2×10^{-8} (air)	Heat exchangers, structural	[85]
Copper Foam (Cu 50-15)	87	0.50	–	–	4.1×10^{-8} (air)	Moderate thickness	[85]
Open-Cell Aluminum Foam	87–95	5–40 PPI (0.17–4)	~25–55	–	1.33×10^{-7} to 1.42×10^{-6}	Experimental airflow tests	[86]
Open-Cell Aluminum Foam	84.5–90.2	1.28–2.89 (8–20 PPI)	36–45	~240	$3.3 \times 10^{-8} – 1.0 \times 10^{-7}$	PV panel cooling	[87]
Open-Cell Foam (Aluminum/Copper)	85–95	0.5–2.5	35–210	–	$10^{-9} – 10^{-7}$ (varies by foam)	PV cooling, heat exchangers	[88]
Open-Cell Copper Foam	97	10, 20, 30, 40, 60 PPI	~210	~350–400	$1 \times 10^{-8} – 3 \times 10^{-7}$	PV cooling, heat exchangers	[82]
Open-Cell Aluminum Foam	97	5, 10, 20, 40 PPI	~30–60	–	$10^{-7} – 10^{-6}$	PV panel cooling	[89]
Open-Cell Aluminum Foam	91.6–94.0	10–30 PPI	4.5–10.5	156–227	$1 \times 10^{-8} – 5 \times 10^{-8}$	Structural, thermal fins	[90]

becomes increasingly essential. The open-cell porous structure allows radiation to penetrate deeper into the material, effectively increasing the heat-transfer surface area. The radiation heat transfer depends on the foam's porosity and pore size distribution, with void spaces reducing conduction but allowing enhanced radiative flux through the foam [94]. The coupling of conduction and radiation complicates the overall heat transfer analysis, requiring advanced models and simulations to predict performance accurately [95].

Numerical simulations and unit-cell models have been widely used to analyze heat transfer in metal foams. These studies provide insight into how pore morphology and flow dynamics affect conduction and convection, enabling optimization of foam structures for maximum thermal efficiency [96]. Experimental studies confirm that metal foam heat sinks outperform traditional designs by virtue of their increased surface area and enhanced convective capabilities, especially when tested with fluids such as air or water under various flow conditions [97].

In PV passive cooling applications, metal foam fins leverage these heat transfer mechanisms to enhance thermal management significantly. Their inherently high thermal conductivity, combined with a large porous surface area, promotes effective natural convection and conduction, resulting in a substantial reduction in PV panel operating temperatures. For instance, copper and aluminium metal foams have been shown to lower panel temperatures by up to 13.2 %, directly improving electrical efficiency by approximately 4–5 % and prolonging the lifespan of PV panels [15,98]. The foam's porosity and pore density critically influence this performance, where an optimal balance maximizes airflow penetration and heat exchange [99]. Furthermore, novel fin configurations such as corrugated metal foam fins have demonstrated thermal effectiveness efficiencies approaching 86 %, outperforming conventional fin designs [93]. These features position metal foam fins as highly promising, sustainable, and low-maintenance components for improving the thermal regulation and operational efficiency of PV panels.

2.2.3.3. Material properties impacting thermal performance. The thermal performance of metal foam fins for PV cooling is strongly governed by key material properties, including porosity, pore size, and the base metal's intrinsic thermal conductivity. High porosity levels (typically 0.90 to 0.95) improve convective heat transfer by enabling greater airflow through the foam structure, facilitating efficient heat extraction from the PV panel surface while maintaining a low pressure drop. For instance, copper metal foams with porosity around 0.94 have demonstrated significant reductions in panel temperature and improvements in electrical efficiency [15,87]. Additionally, smaller pore sizes or higher pore densities increase the effective heat transfer surface area and enhance solid-phase conduction, resulting in better thermal conductivity and heat dissipation [100].

The base metal material critically affects thermal conductivity and overall cooling performance. Copper foams provide superior thermal conductivity, leading to enhanced heat dissipation, while aluminium foams offer a lighter alternative with a good balance between heat transfer and acceptable pressure drops [15,101]. Moreover, compressing metal foams to reduce porosity has been shown to increase the ETC by up to 10-fold due to improved contact between the solid ligaments [102]. However, decreasing porosity also increases foam stiffness and thermal contact resistance, which may impact mechanical compliance and interface performance [103].

Beyond material properties alone, integrating metal foam fins with complementary cooling technologies, such as PCMs, can further enhance thermal management by combining conductive and latent heat transfer mechanisms [104]. Functionally graded metal foams, which exhibit spatial variations in porosity, promote improved temperature uniformity and thermal efficiency, thereby increasing the reliability and effectiveness of PV cooling systems [105]. Consequently, optimizing material porosity, pore size, base metal thermal conductivity, and synergistic design integration is vital to maximize the cooling performance of metal foam fins in PV applications.

2.2.4. Governing equations for convective cooling in metal foam structures

The accurate modelling of convective cooling in metal foam structures relies on fundamental governing equations that describe fluid flow and heat transfer within the porous matrix. Due to the complex and irregular geometry of metal foams, porous media theory is often applied to represent the macroscale behavior by averaging microscale phenomena [98,106]. This approach simplifies complex flow paths into effective properties, such as porosity and permeability, which are critical for capturing convective cooling performance.

The continuity equation governs mass conservation in the fluid phase, commonly assuming incompressible and steady flow conditions within the porous medium [107]:

$$\nabla \cdot u = 0 \tag{10}$$

where u denotes the fluid's velocity vector at the surface. This equation ensures fluid mass balance and is fundamental in describing fluid transport through the foam's porous network [108].

To describe momentum conservation, the extended Darcy model, known as the Brinkman-Forchheimer equation, is widely adopted. This model incorporates viscous, inertial, and permeability effects, effectively bridging laminar to moderate flow regimes observed in metal foams [109]:

$$\mu \nabla^2 u - \frac{\mu}{K} u - \rho C_F |u| u + \nabla p = 0 \tag{11}$$

Here, μ represents the fluid dynamic viscosity, K is the foam permeability, ρ is fluid density, C_F is the inertial resistance (Forchheimer) coefficient, and p is pressure. This equation captures the flow resistance and pressure loss caused by the intricate porous structure of metal foams [109].

Heat transfer in metal foams involves simultaneous conduction through the solid skeleton and convection through the fluid. The coupled energy conservation equation for the solid and fluid phases in the porous medium is expressed as [110]:

$$(1-\epsilon)\rho_s c_{p,s} \frac{\partial T_s}{\partial t} + \epsilon \rho f c_{p,f} \frac{\partial T_f}{\partial t} = \nabla \cdot \left(k_{eff} \nabla T\right) + h_{sf} a_s \left(T_f - T_s\right) \tag{12}$$

Where ϵ is porosity, ρ and c_p are the density and specific heat capacity of a solid (s) and fluid (f) phases respectively, T is temperature, k_{eff} denotes effective thermal conductivity, h_{sf} is the interfacial heat transfer coefficient, a_s represents the foam's specific surface area. This equation effectively couples conduction within the foam's ligaments and convective heat exchange with the fluid flowing through the pores [110]. The ETC k_{eff} is a critical parameter influencing thermal performance, accounting not only for conduction in the solid phase but also for radiative heat transfer within the porous matrix. It varies with porosity, temperature, and foam morphology, requiring accurate characterization for reliable thermal simulations [111].

Advanced numerical simulations, often employing Computational Fluid Dynamics (CFD), use these governing equations to predict convective cooling efficiency in metal foam heat sinks and fins under various operating conditions. Analytical and semi-empirical models based on unit-cell geometries and pore-scale flow also complement numerical approaches, facilitating the design optimization of metal foam structures for improved thermal management [78,112]. However, the governing equations for convective cooling in metal foam structures integrate fluid flow and heat transfer dynamics through porous media theory. Understanding and accurately modelling these equations are essential for developing high-performance metal foam fins and passive cooling solutions, including those targeting PV panel thermal regulation.

3. Combined passive cooling approaches: Radiative paints and metal foam fins

This section explores the integration of radiative paint coatings and metal foam fins as a hybrid passive cooling strategy for PV panels. It synthesizes current knowledge of the underlying thermal mechanisms, critically evaluates experimental and field performance data, compares this hybrid approach with conventional cooling methods, and highlights practical considerations for implementation and scalability.

3.1. Synergistic thermal mechanisms of radiative and convective cooling

Radiative paints and metal foam fins have independently demonstrated significant potential as passive cooling techniques for PV panels by leveraging spectral heat emission and enhanced convective heat transfer, respectively. Radiative cooling functions by emitting infrared radiation within the atmospheric transparency window (8–13 μm), allowing PV panels to dissipate heat to the cold outer space without energy consumption, thereby reducing operating temperatures by up to 10 °C and improving efficiency and lifespan [113]. Conversely, metal foam fins increase the effective surface area for heat exchange and promote airflow through their porous open-cell structures, resulting in substantial temperature reductions and enhanced convective cooling performance [98,114].

Despite these proven benefits, no experimental studies have yet explored or reported the combined use of radiative paints and metal foam fins for PV cooling. This represents a critical knowledge gap and a promising avenue for multidisciplinary research to develop hybrid passive cooling systems that maximize thermal regulation without external energy inputs. Theoretically, integrating radiative paints with metal foam fins could synergistically enhance PV thermal management by combining two complementary mechanisms. Radiative coatings applied to the PV front or back surface would facilitate efficient infrared emission, reducing the panel's surface temperature, particularly under high solar irradiance and low convective conditions [22,115]. Simultaneously, metal foam fins attached to the panel's rear or edges would enhance natural convection by increasing airflow and disrupting thermal boundary layers, thereby improving heat dissipation from the module [116]. This dual cooling pathway is expected to achieve temperature reductions greater than those attainable by either method alone, potentially leading to higher power conversion efficiencies and extended module lifespans.

Challenges to this combined approach include ensuring material compatibility to prevent degradation, avoiding obstruction of radiative emission by foam placement, and optimizing foam porosity and geometry to balance airflow resistance and heat transfer efficiency [117]. Environmental factors such as humidity, dust accumulation, and UV exposure may also impact long-term performance and durability, necessitating comprehensive testing under realistic outdoor conditions.

The potential impact of such hybrid systems is considerable, especially for PV installations in off-grid, arid, or harsh environments where active cooling is impractical or cost-prohibitive. The energy-free, maintenance-light nature of combined radiative and convective passive cooling could substantially improve the economic viability and reliability of solar power systems globally. To guide future research, we propose a conceptual schematic illustrating the complementary heat transfer mechanisms: radiative emission from coated surfaces paired with enhanced convective airflow facilitated by porous metal foam fins, as shown in Fig. 10. This schematic serves as a foundational model for experimental validation and coupled thermal-fluid dynamic simulations aimed at optimizing system design and materials selection.

3.2. Critical evaluation of experimental and field performance data

Despite extensive research on radiative paints and metal foam fins as standalone passive cooling techniques for PV panels, no published experimental or field studies currently investigate their combined application for PV cooling. This absence represents a significant knowledge gap and a promising opportunity for future research.

Metal foam fins have demonstrated considerable effectiveness in reducing PV panel temperatures and improving power output. For example, open-cell copper metal foam fins reduced average PV panel

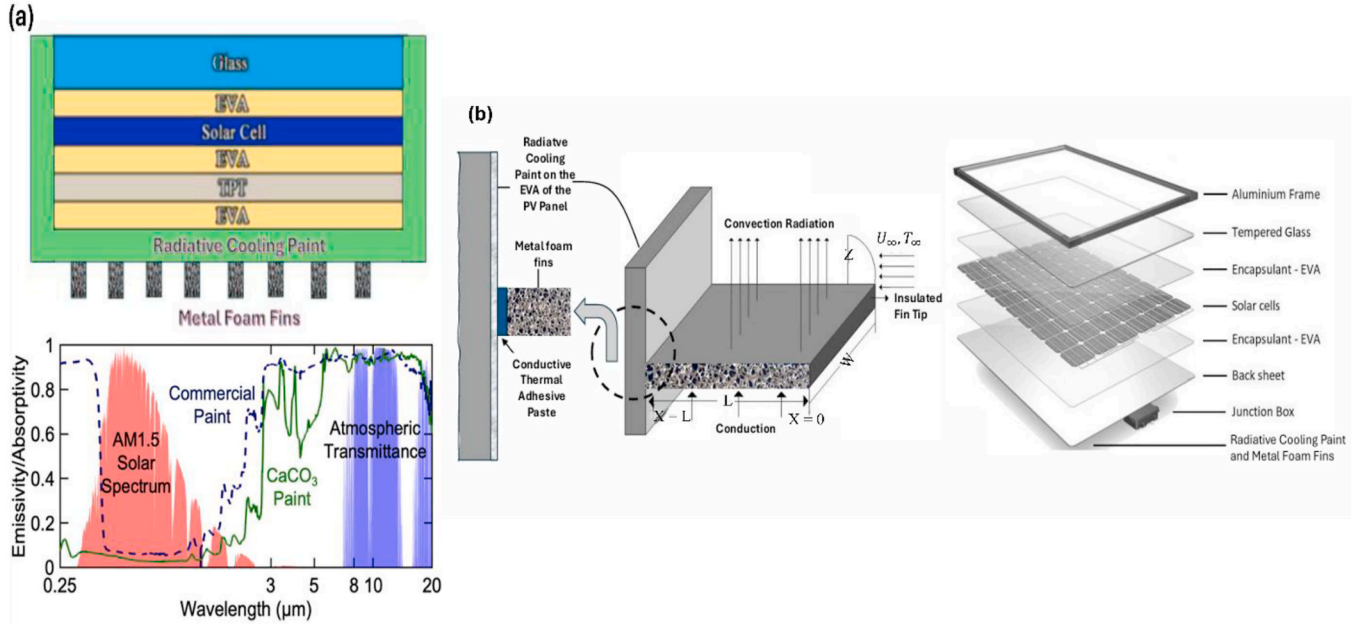


Fig. 10. Conceptual schematic of synergistic passive cooling integrating radiative paint and metal foam fins on PV panels. (a) Cross-sectional view showing radiative paint applied on the EVA layer and metal foam fins attached to the rear surface. The spectral emissivity and absorptivity of the radiative paint are compared with those of commercial paints and with atmospheric transmittance within the 8–13 μm infrared window. (b) Thermal management schematic highlighting conduction, convection, and radiation processes in the hybrid system, with an exploded view illustrating the physical integration and complementary cooling roles of both components.

temperature by approximately 8.4 %, resulting in a 4.9 % increase in power output [15]. Similarly, aluminium-based metal foams tested in compact heat exchangers for concentrating PV receivers showed improved heat transfer and pressure drop characteristics, potentially increasing electricity production by about 1.5 % [118]. Various fin geometries and porosities have been shown to influence cooling efficiency positively [119].

Radiative cooling paints utilize thermal emission within the atmospheric transparency window (8–13 μm) to dissipate heat effectively, lowering PV operating temperatures by up to 10 °C and enhancing module efficiency and lifespan [22,113]. Coatings such as PDMS and polyvinyl butyral (PVB) have been studied for their radiative properties, with PVB showing superior transparency and cost advantages [120]. Radiative cooling methods provide daytime cooling even under direct sunlight, addressing key challenges in thermal management [121]. While the theoretical benefits of combining radiative paints and metal foam fins are promising, the absence of experimental validation highlights challenges related to material compatibility, optimal integration, and durability under environmental conditions. Addressing these practical issues through targeted laboratory and field investigations is crucial to advancing this hybrid cooling approach [122].

Fig. 11 presents a comparative visualization of recent experimental findings on temperature reduction and efficiency improvement achieved through these passive cooling methods. Copper and aluminium metal foam fins demonstrate substantial temperature reductions (8.4 °C and 9.2 °C, respectively) alongside meaningful efficiency gains (4.9 % and 8.9 %, respectively). Radiative cooling using double glazing and radiative coatings (PDMS, PVB) also contributes significant cooling and efficiency benefits, with temperature reductions up to 10 °C and efficiency improvements exceeding 14 %. This comparative data highlights the potential performance enhancements each method offers individually, reinforcing the value of exploring their combined application.

Given the lack of experimental and field validation of this combined approach, targeted laboratory studies and pilot-scale outdoor testing are urgently needed. Such investigations would clarify the practical benefits, integration strategies, and cost-effectiveness of hybrid passive cooling systems, potentially enabling low-cost, energy-free thermal management solutions that extend PV module lifespan and improve energy yield across diverse climates.

3.3. Comparative performance analysis with existing cooling methods

PV panel cooling techniques vary in their mechanisms, effectiveness, cost, and maintenance demands. Passive methods such as radiative paint and metal foam fins are gaining attention due to their ability to reduce operating temperatures without consuming additional energy or requiring active maintenance.

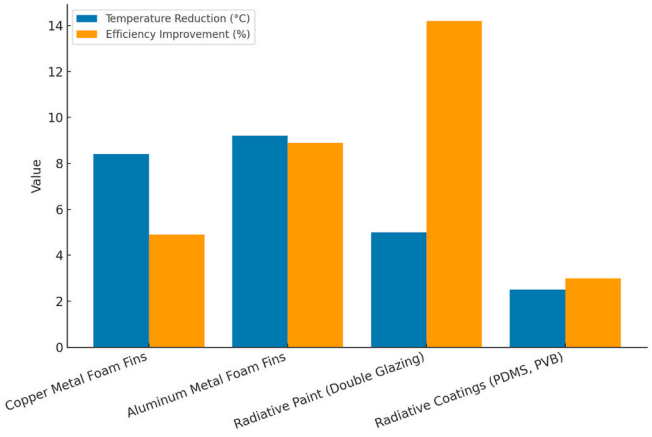


Fig. 11. Comparative performance of different passive cooling methods for PV panels: temperature reduction (°C) and efficiency improvement (%) (Data sources: Hasan & Farhan [15], Hamdan et al. [123], Shabeeb et al. [124], Goodarzi et al. [120]).

Radiative paint increases the thermal emissivity, allowing PV panels to reject heat via infrared radiation, generally resulting in temperature drops of the order of 6.6 °C and efficiency gains of the order of 11 %. The technique is most effective in clear-sky, dry climates, but is extremely sensitive to climate conditions. Metal foam fins, by contrast, increase natural convection by presenting a high-surface-area matrix for heat rejection. Such fins can lower PV temperatures by as much as 12 °C, with varying levels of efficiency improvement reported. They must be precisely manufactured but provide solid, energy-free cooling. Table 4 provides an overview of the performance and viability of these cooling approaches and other available passive solutions, such as PCMs, finned heat sinks, perforated aluminium plates, and pin-finned copper sheets.

The expected synergy of combining radiative paint with metal foam fins offers a hybrid passive cooling system that exploits both radiative and convective heat transfer mechanisms. Although current performance estimates of these components are derived from separate studies, their synergy offers the potential for greater temperature reductions and efficiency improvements than either method alone, without entailing energy or maintenance costs. It is particularly promising for stand-alone or low-maintenance PV systems in climates that favor radiative cooling and natural convection. Experimental confirmation of the integrated system is still pending, however. Future efforts must be directed towards empirical testing to validate the theoretical benefits and the practical performance of this hybrid system. This comparative analysis underscores the value of exploring innovative passive cooling techniques that balance efficiency, simplicity, and cost, potentially expanding the options for sustainable PV thermal management.

3.4. Practical considerations for implementation and deployment

The application of radiative paints and metal foam fins in a hybrid passive cooling system for PV panels raises a series of practical concerns and implementation challenges. Material compatibility is one of the first main concerns. The radiative paints should be capable of good adhesion to the metal foam fins without compromising their thermal emissivity and long-term durability [15,134]. Additionally, thermal expansion differences between the paint and metal foams must be addressed to ensure structural stability under fluctuating temperature conditions.

Design optimization of the foam fin geometry and placement relative to the radiative coating is crucial. The hybrid system must be carefully engineered to enhance convective heat transfer without impeding radiative cooling. Fins, for instance, must be situated such that air can pass through the metal foam while infrared radiation from coated surfaces is not impeded [135]. Effective incorporation of these elements, either retrofitted into current panels or engineered into new production, demands diligent design and system planning.

Environmental considerations also play a pivotal role. Dust accumulation, humidity, and UV exposure can influence the long-term efficiency and maintenance needs of this hybrid cooling system. Therefore, materials must be chosen for their resilience against such environmental factors to ensure reliable long-term performance [136]. A focus on low-maintenance, durable solutions could enhance the economic feasibility of this passive system, particularly for off-grid or low-maintenance PV installations in arid or harsh environments.

Furthermore, scalability is a significant consideration. The ability to adapt this hybrid cooling system to different scales, from residential to large-scale commercial PV farms, will determine its overall feasibility. The system's cost-effectiveness will also depend on the scalability of the manufacturing, especially considering that the integration of nanomaterials or complex photonic structures can entail higher production expenses [41,137]. Continued research in cost reduction and advanced manufacturing methods will be key to making this system viable for general use.

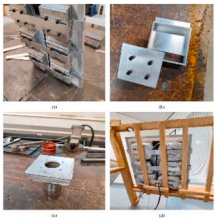
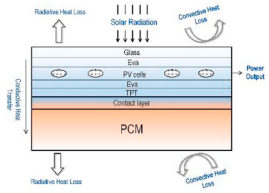
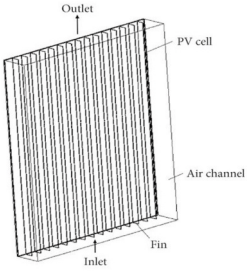
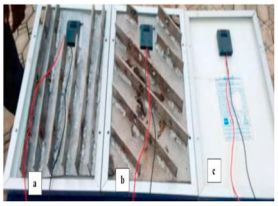
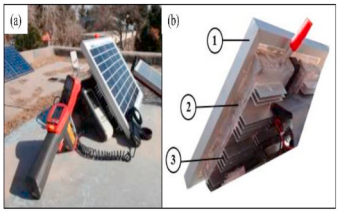
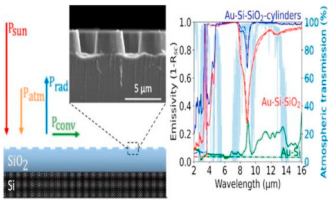
In summary, while integrating radiative paints and metal foam fins into a hybrid passive cooling system offers great promise, it requires careful attention to material compatibility, design optimization, and environmental adaptability. Additional investigation into the scalability, affordability, and long-term effectiveness of such systems is needed to fully harness their potential as cost-effective and sustainable solutions for PV cooling.

3.5. Quantitative analysis of the hybrid system

3.5.1. Thermal resistance distribution model

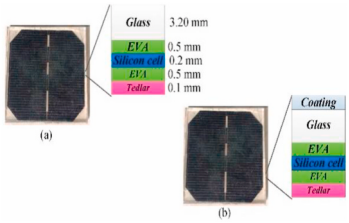
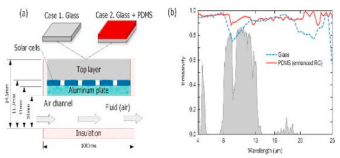
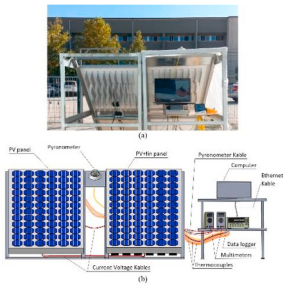
The hybrid radiative–convective cooling system is modelled using an equivalent thermal-resistance network that captures the coupled radiative, conductive, and convective mechanisms. The total equivalent resistance, expressed as a combination of serial and parallel elements as

Table 4
Comparative overview of passive cooling techniques for PV panels.

Cooling Method	Configurations/Setup	Typical Temperature Reduction (°C)	Efficiency Improvement (%)	Advantages	Limitations	Reference (s)
PCM		10	5.23	- Effective cooling by absorbing latent heat - Simple to implement with a small container system	- PCM performance depends on environmental conditions - Inappropriate PCM can reduce efficiency (melting issues)	[125]
		23	5.18	- Effective cooling by absorbing latent heat - Simple implementation with small container systems	- PCM performance varies with environmental conditions - Inappropriate PCM selection may reduce efficiency (e.g., melting issues)	[126]
Heat Sinks (with fins)		5–7	6–8	- Simple and cost-effective. - Enhanced heat dissipation via increased surface area. - No additional energy required.	- The cooling effect is limited by ambient temperature and airflow conditions.	[127]
		2.48–5	6.5–6.8	- Simple and cost-effective. - Enhanced heat dissipation via increased surface area. - No additional energy required.	- Cooling effect depends on ambient temperature and solar irradiance.	[128]
		13.08–16.34	13.5	- Enhanced cooling performance with nanofluid. - Improved heat transfer properties due to aluminium nanoparticles.	- Long-term stability of nanofluid is a concern.	[129]
Radiative Cooling (Silica Microcylinders)		8	_	Enhanced radiative cooling. - Passive cooling, no energy input required.	- Manufacturing challenges for large-scale integration.	[130]

(continued on next page)

Table 4 (*continued*)

Cooling Method	Configurations/Setup	Typical Temperature Reduction (°C)	Efficiency Improvement (%)	Advantages	Limitations	Reference (s)
Polymeric Coatings		3–6	3.6–4	- Cost-effective and easy to apply. - Passive cooling without energy input.	- Long-term durability of the coating is still under evaluation.	[131]
Radiative cooling		1.7	0.76	- Efficient radiative cooling. - Passive cooling.	- Limited by environmental factors like low light.	[132]
Aluminum Fins		3.39	11.55	- Simple, cost-effective, and improves power output.	- Cooling effect is limited without external airflow; modest efficiency improvement.	[133]

illustrated in Fig. 12, quantifies how heat from the PV surface is dissipated through both the radiative coating and the metal-foam fin layer [138].

$$R_{tot}^{-1} = R_{rad}^{-1} + (R_{cond} + R_{int} + R_{conv})^{-1} \tag{13}$$

where.

R_{rad} represents the radiative thermal resistance of the coating surface,

R_{cond} denotes the conduction resistance through the PV–foam contact region,

R_{int} is the interfacial contact thermal resistance at the PV–foam interface, and

R_{conv} corresponds to the convective thermal resistance within the porous foam structure.

Each resistance term can be expressed as [139]:

$$R_{rad} = \frac{1}{\varepsilon_{eff}\sigma A\left(T_s^2 + T_{amb}^2\right)(T_s + T_{amb})}, R_{conv} = \frac{1}{h_{eff}A}, R_{int} = \frac{\delta_{int}}{k_{int}A} \tag{14}$$

Where ε_{eff} is the effective emissivity of the coated surface, σ is the Stefan–Boltzmann constant, A is the effective radiating area, T_s and T_{amb} are the surface and ambient temperatures, respectively, and h_{eff} the effective convective coefficient of the metal-foam structure.

As shown in Fig. 12, heat from the PV surface is distributed between a radiative path (R_{rad}) and a series path ($R_{cond} + R_{int} + R_{conv}$). The total heat-flow rate is [139]:

$$\dot{q}_{tot} = \frac{\Delta T}{R_{tot}}, \Delta T = T_s - T_{amb}. \tag{15}$$

A smaller R_{tot} indicates improved overall heat-dissipation capability. This analytical framework provides the basis for quantifying the combined interaction between the radiative and convective mechanisms.

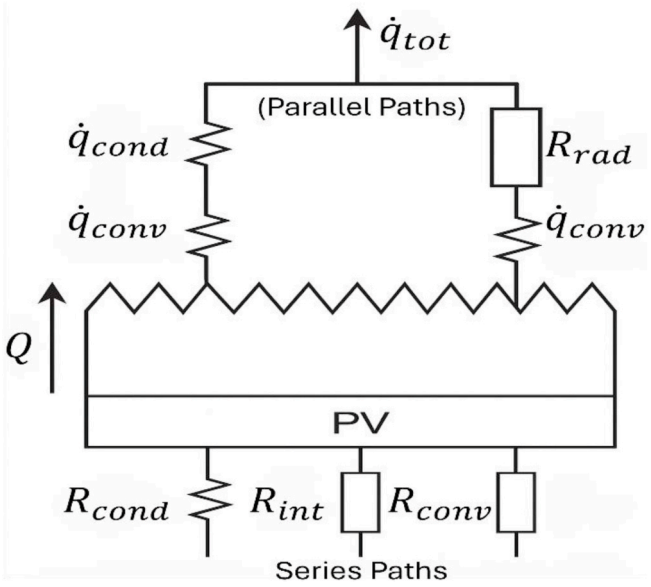


Fig. 12. Thermal resistance network of the hybrid radiative–convective cooling system.

3.5.2. Quantitative evaluation of system performance

Based on the equivalent-resistance model established in Section 3.5.1, the total heat flux from the PV surface is governed by the overall thermal resistance R_{tot} which couples radiative and convective heat-transfer mechanisms. When both dissipation paths contribute comparably, the surface temperature decreases significantly, improving PV efficiency.

Table 5
Cost structure and economic considerations of radiative paint and metal-foam fin components for hybrid passive PV cooling systems.

Cost component	Radiative paint	Metal foam fins
Material costs	- Generally low cost due to simple preparation and use of inexpensive materials like glass bubbles [142]. - Some formulations use recycled materials, further reducing costs [143]. - Water-based paints are cost-effective and environmentally friendly [74,144].	High cost due to the expensive manufacturing process and materials [145, 146]. - Costs remain high despite improvements in quality [147].
Production costs	- Low production costs due to scalable and simple manufacturing processes [122,148]. - Use of standard materials and straightforward mixing methods [142].	- High production costs due to complex manufacturing processes [149]. - Requires specialized equipment and labour [145].
Application costs	- Easy and low-cost application similar to conventional paints [150]. - Can be applied to various surfaces without specialized equipment [122].	- Higher application costs due to the need for precise installation and integration into systems [151]. - Requires skilled labour for installation [152].
Maintenance Costs	- Low maintenance costs due to durability and weather resistance [153]. - Some formulations offer self-cleaning properties [150].	- Potentially high maintenance costs due to susceptibility to damage and wear [154]. - Requires regular inspection and potential replacement [151].
Economic Viability	- High economic viability due to low costs and ease of application [155]. - Suitable for large-scale applications [155].	- High costs limit economic viability [145,146]. - More viable in applications where multiple functions are needed [147].

(a) Parameter matching between coating and foam

For optimal performance, the two branches should be thermally balanced such that [140]:

$$R_{rad} \approx R_{cond} + R_{int} + R_{conv} \tag{16}$$

This condition prevents one mechanism from dominating heat transfer. R_{rad} is primarily affected by the coating's emissivity (ε_{eff}) and spectral selectivity, while R_{cond}, R_{int}, *and* R_{conv} depend on the foam's porosity, thickness, and thermal conductivity. Proper parameter matching minimizes R_{tot} and enhances cooling efficiency.

(b) Influence of foam-fin geometry and interface contact

The geometric configuration of the metal-foam layer affects both the optical emission and convective flow characteristics. When the foam is placed too close to the radiative coating, partial blockage of infrared emission increases R_{rad}; excessive spacing, however, weakens natural convection and raises R_{conv}. The optimum separation distance d_{opt} that minimizes the total thermal resistance satisfies [141]:

$$\frac{dR_{tot}}{dd} = 0 \; at \; d = d_{opt} \tag{17}$$

The interfacial contact resistance R_{int} also plays a crucial role. Imperfect bonding or trapped air gaps increases R_{int}, whereas improved mechanical adhesion and thermally conductive interfaces lower this resistance, enhancing total heat removal.

(c) Quantitative synergistic effect

The collective behavior between the radiative and convective cooling mechanisms can be quantified using a synergistic-efficiency factor:

$$\eta_{syn} = \frac{\dot{q}_{tot} - (\dot{q}_{rad} + \dot{q}_{conv})_{indiv}}{(\dot{q}_{rad} + \dot{q}_{conv})_{indiv}} \times 100\% \tag{18}$$

Where $(\dot{q}_{rad} + \dot{q}_{conv})_{indiv}$ represents the combined heat flux from the individual radiative and convective mechanisms when acting independently, without interaction or coupling effects.

A positive η_{syn} indicates that the hybrid configuration provides superior cooling performance than the sum of the individual radiative and convective effects, confirming the presence of a synergistic interaction between the two heat-transfer modes.

3.6. Techno-economic analysis

3.6.1. Cost components

The techno-economic feasibility of hybrid passive cooling systems for PV modules depends strongly on the cost structure of their core components, namely radiative paints and metal foam fins. Each technology presents a distinct cost profile shaped by differences in material composition, fabrication processes, and deployment methods. Radiative paints benefit from inexpensive, readily available ingredients and scalable production processes, whereas metal foam fins require complex manufacturing and higher capital requirements. A clear understanding of these cost elements provides the foundation for subsequent payback and scalability analyses. The comparative cost structure is summarized in Table 5.

Overall, the results highlight that radiative paint systems demonstrate superior cost efficiency across nearly all stages, from material sourcing to application and maintenance, making them suitable for broad industrial implementation. Metal foam fins remain cost intensive due to specialized manufacturing and integration requirements, although their thermal advantages may justify use in high-performance or multifunctional applications. This comparative insight emphasizes the importance of balancing thermal gains with cost competitiveness in advancing large-scale passive PV cooling solutions.

3.6.2. Cost per unit area and payback period

The system cost per unit area for passive PV cooling technologies is determined by the combined material, application, and maintenance costs of each component. For radiative paints, expenses arise from pigments such as $BaSO_4$, $CaCO_3$, and hBN that provide high solar reflectance and emissivity, along with application and environmental-compliance costs associated with low-VOC formulations [156,157]. For metal-foam cooling, cost factors are mainly influenced by the porosity, pore density, and filling ratio of the copper metal foam, which

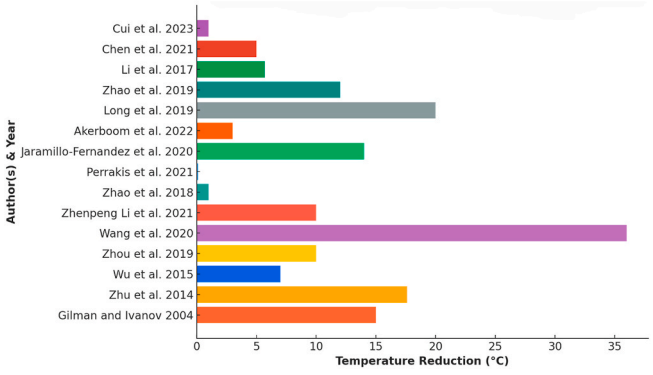


Fig. 13. Temperature reduction through various cooling techniques for PV panels. This illustrates the temperature reductions achieved by various passive cooling techniques, with studies over the years indicating the performance of each method in lowering panel temperatures.

determine both thermal performance and material use. Studies indicate that a 75 % filling rate achieves an effective balance between heat transfer efficiency and production cost, maintaining temperature uniformity while substantially reducing foam consumption and fabrication effort [158,159]. The payback framework links these costs to thermal and electrical benefits through a simple ratio [160]:

$$Payback\ period\ (years) = \frac{System\ cost\ per\ area\ (US\$m^{-2})}{Annual\ energy\ saving\ per\ area\ (kWhm^{-2}yr^{-1}) \times Electricity\ price\ (US\$kWh^{-1})} \quad (19)$$

Temperature reduction directly governs energy savings because PV efficiency typically decreases by 0.4–0.5 % per °C rise in temperature [161]. Thus, a 6 °C cooling translates to roughly a 2.4–3 % efficiency gain, which can be converted to extra annual energy yield using site irradiance data or simulations such as the System Advisor Model (SAM) [162].

When precise site information is unavailable, representative literature values provide reasonable estimation bounds. Reported hourly global horizontal irradiance values typically reach 300–650 W/m^2 at mid-day in clear-sky conditions, depending on location and season. Annual average GHI may be around 150–200 W/m^2 for some regions [163]. Residential or commercial electricity tariffs often range from US $0.10–0.40 per kWh, depending on country and market conditions. Cooling strategies delivering 5–10 °C temperature drops yield 2–5 % efficiency improvements and can shorten payback to 4–8 years under mid-range tariff scenarios [164]. This simplified yet robust framework therefore connects the physical temperature reduction of radiative paints and metal-foam fins directly to their economic viability, providing a practical tool for comparing passive and hybrid PV-cooling systems under varied operating conditions.

3.6.3. Scalability and cost reduction pathways

Large-scale fabrication of radiative paints and metal-foam fins for PV cooling can be achieved through industrially scalable techniques such as roll-to-roll coating, spray deposition, and additive manufacturing, which enable continuous, high-throughput, and low-cost production on large substrates [165]. Metal foams can also be manufactured economically at scale using pressurized infiltration of molten metals into salt preforms or self-propagating combustion synthesis, both allowing precise control of pore structure and porosity [166,167]. Material costs can be minimized by incorporating abundant or recycled resources such as glass bubbles and waste polystyrene, which retain high solar reflectivity and emissivity while reducing overall expenses [142]. Furthermore, integrating automation and modular assembly within coating and foam production lines lowers labor and installation costs, while economies of scale from mass manufacturing further reduce the cost per unit area. Collectively, these strategies establish practical pathways toward cost-effective, scalable deployment of radiative paint and metal-foam fin cooling technologies for enhancing PV module performance.

3.6.4. Comparative industrial feasibility

Radiative paints and metal foam fins represent two promising passive cooling technologies for PV modules, offering distinct cost-performance characteristics. Radiative cooling paints can achieve solar reflectance up to 98.1 % and emissivity of 0.96, delivering cooling powers of 117 W/m^2 and temperature reductions exceeding 4.5 °C below ambient during field operation [157]. Although fluoropolymer-based paints are about five times more expensive than conventional acrylic coatings, advanced bilayer designs reduce material use by 70 %, lowering the overall cost to about US $1.3 m^{-2} [168]. Furthermore, recent water-based and low-VOC formulations enhance environmental compatibility and scalability for large-area PV deployment [156]. In contrast, metal foam fins offer superior thermal uniformity and improved efficiency, with studies reporting improvements in temperature uniformity of 27.08 %, electrical efficiency gains of up to 6.18 %, and overall thermal efficiency increases of 59.53 % [158]. However, their high material and fabrication costs limit economic viability, though optimizing the 75 % filling ratio can significantly reduce costs while maintaining thermal performance.

When compared with existing PV cooling systems, such as forced-air aluminium-fin configurations that achieve surface-temperature reduction of 14.9 % and output-power improvement of 6.7 % with a payback period of about 5.1 years [169,170], radiative paints exhibit a better balance between cost, simplicity, and scalability. Metal foam fins, though cost-intensive, provide high-performance stability for hybrid systems under extreme irradiance. Overall, radiative paints are more economically and environmentally favorable for large-scale PV integration, while metal foam fins are an effective enhancement in hybrid configurations that require superior temperature regulation and efficiency.

4. Key performance metrics and evaluation methods

In order to evaluate the performance of passive cooling systems for PV panels, there is a need to define appropriate performance criteria and test methods. This section identifies the most critical parameters to assess during testing of cooling methods, including temperature reduction, energy efficiency, durability, and scalability.

4.1. Temperature reduction and cooling efficiency

Temperature reduction is among the most significant factors for increasing the lifespan and efficiency of PV panels. Passive cooling systems reduce the surface temperature of PV panels, which is essential for delivering maximum performance. The performance of these cooling techniques is generally measured in terms of surface temperature drop (ΔT) and cooling capacity (W/m^2), which quantify the decrease in temperature and the heat removed by the system, respectively. Fig. 13 shows the cooling performance of different passive methods implemented on various types of PV panels.

Several studies have demonstrated the potential of different passive cooling strategies. Gilman and Ivanov [171] employed quantum-assisted emission as the cooling technique in PV panels and reported a temperature drop of 5–20 °C with an efficiency increase of 3–10 %. The method demonstrates the potential of novel emission-based coatings to lower the surface temperature of PV panels effectively. Similarly, Zhu et al. [172] deposited a micro-photonic structure on crystalline silicon wafers and achieved an enormous temperature drop of 17.6 °C with efficiency enhancement by 7.9 %. Their research highlights the potential of photonic structures to cool PV panels by enhancing their optical properties.

In another study, Wu et al. [173] used a benzocyclobutene coating, resulting in a 7 °C reduction in temperature and a 0.5 % increase in efficiency. This demonstrates a modest yet significant improvement in cooling efficiency. Zhou et al. [113] used a soda-lime glass-based radiative cooler on concentrated PV modules, reducing module temperature by 10 °C, increasing open-circuit voltage by 5.7 %, and extending module lifespan by 40 %. Similarly, Wang et al. [42] applied soda-lime

glass coolers on both sides of the heat sink for GaSb solar cells, achieving temperature reductions of 5–36 °C and an 8–27 % increase in open-circuit voltage.

The application of PDMS coatings has also been widely explored. Zhenpeng Li et al. [174] applied a 200 μm PDMS layer to PV panels, leading to a temperature reduction of approximately 1 °C for insulated PV panels, and up to 10 °C for ventilated PV panels. This study highlights the insulating effect, which provides better temperature control in ventilated PV panels. Similarly, Zhao et al. [175] used a 200–300 μm PDMS layer, achieving an average temperature reduction of 1 °C, with slight variations based on environmental conditions.

Perrakis et al. [176] studied square micro-gratings with nanopillars on Si wafers, leading to a temperature reduction of 5.8 °C for Si wafers, though the effect was less pronounced for commercial PV panels (0.1 °C). Jaramillo-Fernandez et al. [177] used spheres with particle sizes ranging from 75 to 250 μm, resulting in a temperature reduction of 14 °C. Their findings indicated that smaller particle sizes (75 μm) exhibited better optical properties and were more effective at reducing temperatures.

Zhu et al. [47] applied a square lattice of silica pyramids on PV panels, resulting in a temperature reduction of 17.6 °C, demonstrating the high optical properties of silica compared to other materials. Akerboom et al. [178] and Long et al. [179] explored microcylinder arrays and silica cylinders on Si wafers, achieving temperature reductions of 3 °C and 20 °C, respectively. Li et al. [180] used multilayer coatings, resulting in a temperature reduction of 5.7 °C, while Chen et al. [54] applied nanospherical coatings, which led to a temperature reduction of 5 °C due to improved transmissivity and reflectivity. Finally, Cui et al. [55,181] investigated multilayer and TPX/SiO_2 coatings and achieved temperature drops of 3 °C and 1 °C, respectively. These studies emphasize the potential of tailored coatings in enhancing the cooling efficiency of PV systems.

These results show that different cooling techniques, from photonic structures and radiative coolers to coatings and multilayer designs, also hold great promise for enhancing the temperature difference and cooling efficiency of PV systems. With continued innovation in cooling technologies, even better performance and sustainability of solar power systems can be expected. Metal foam fins and radiative paints, especially, seem highly promising passive cooling alternatives for PV panels because of their potential to mitigate surface temperatures and thus enhance the efficiency and longevity of PV systems.

Metal foam fins, due to their high surface area and porosity, enhance heat dissipation and help maintain cooler temperatures, especially when hybridized with other methods such as PCMs. Such systems provide effective cooling across a broad range of environmental conditions and can be designed for use with various PV technologies. Likewise, radiative paints in the form of high-solar-reflectance and thermal-emittance coatings, such as $BaSO_4$ and PDMS-based, provide an inexpensive and readily deployable solution. These have been shown to offer temperature drops of 1 °C–10 °C and are suitable for large commercial applications. Used in conjunction with other cooling methods, radiative paints also help achieve optimal thermal regulation for PV panels. Together, these technologies hold tremendous potential to increase energy efficiency and operational lifespan of solar power systems, making way for greener, more affordable solar power generation.

4.2. Energy efficiency and performance improvements

Cooling of PV panels is significant to improve their PCE. The efficiency of PV panels decreases with increasing temperature, with a well-established decrease of about 0.45 % in output power per 1 °C increase above 25 °C. The performance of cooling systems is quantified by key parameters such as energy efficiency gains (Δ Efficiency), improvement in power output, and long-term performance and stability.

Different passive cooling techniques, including radiative paints, metal foam fins, and other new materials, lower the PV panels' surface temperature, thereby enhancing their efficiency and lifespan directly. For instance, water-based and PCM-based hybrid cooling systems reduce the temperature by 20–30 %, which improves the power output by up to 13.31 % [182]. Similarly, passive cooling methods, including the use of PDMS coatings or micro-photonic structures, have reported reductions in temperature by 1 °C–10 °C, demonstrating the promise of such coatings in optimizing PV performance [183].

The temperature drop not only increases short-term power output but also increases the working life of PV systems. Minimizing temperature degradation helps prevent material fatigue and reduce the rate of efficiency loss over time. For instance, research has shown that PV panels that operate at lower temperatures exhibit significantly better long-term stability and lower degradation rates, thereby rendering PV systems economically feasible [184]. The introduction of new cooling solutions, such as thermoelectric generators (TEGs) and automatic cleaning systems, also improves efficiency in the long term [185].

Adding cooling systems to PV panels also enhances overall power output. For example, a hybrid PV/T system using a dimpled teardrop plate exhibited 13.79 % higher maximum power, whereas the PCM-based cooling systems provided 3 % enhancement in power [186]. These results emphasize not only the need to optimize temperature but also to ensure long-term performance and reliability. Long-term performance analysis, i.e., analyzing the degradation rates and using time-series decomposition approaches, provides valuable insights into the effects of cooling methods on the working life of PV modules [187]. The incorporation of these cooling systems has excellent potential to enhance both the energy efficiency and operational lifetime of solar energy systems. As research and development continue to advance cooling materials and systems, the performance characteristics, such as energy efficiency, power output, and long-term stability, will continue to improve, providing an environmentally friendly and cost-effective route to solar energy harvesting in the future, as shown in Fig. 14.

4.3. Durability and longevity

The durability and longevity of passive cooling systems, especially radiative paints and metal foam fins, play a vital role in enhancing the performance and longevity of PV panels. The cooling systems mitigate the negative impacts of excessive heat buildup in PV panels, which may reduce their efficiency and accelerate material degradation. Although both technologies have shown high potential to enhance PV performance, their long-term efficiency and durability remain the most critical factors to be tested for practical large-scale application.

Radiative paints, which reflect solar radiation and emit infrared heat, face degradation from UV radiation, temperature fluctuations, humidity, and dust accumulation. UV exposure leads to discoloration and reduced reflectivity, while dust accumulation can diminish their cooling performance [156]. Materials such as TiO_2 and $BaSO_4$ are commonly

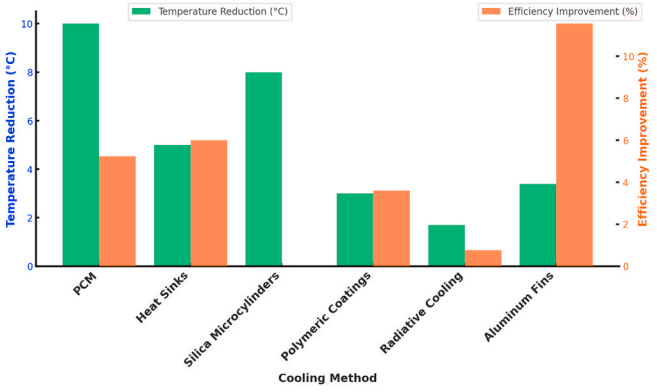


Fig. 14. Efficiency improvement and temperature reduction for different passive cooling methods in PV panels (Data source: [125,127,130–133]).

used for their high reflectivity and durability, but their long-term effectiveness can be compromised by environmental pollutants [15, 94]. Table 6 provides a quantitative comparison of the climatic adaptability of various radiative coatings, highlighting their durability under UV, humidity, and dust exposure. These indicators offer insight into the environmental resilience of radiative materials and their suitability for long-term PV deployment. Metal foam fins, typically fabricated from aluminium or copper, enhance convective heat dissipation due to their porous structure. Nonetheless, they are vulnerable to corrosion, particularly in humid or saline environments, which can compromise thermal conductivity and mechanical stability over time [118].

Key material properties such as thermal emissivity, solar reflectance, and porosity directly influence the longevity of both cooling systems. For radiative paints, high solar reflectance and thermal emissivity are crucial for effective cooling. The longevity of metal foam fins depends on optimizing porosity and ensuring corrosion resistance, as these properties determine their efficiency and durability [193]. Hydrophobic coatings can extend the life of the systems by minimizing the effect of environmental contaminants [156].

Radiative paints and metal foam fins have a synergistic cooling effect together. As radiative paints reflect solar radiation, metal foam fins enhance convective heat dissipation, resulting in greater temperature drops and panel efficiency. This two-pronged methodology can significantly improve the performance and durability of PV panels [22,72]. There are, however, difficulties in combining these materials, such as achieving high paint adhesion on the foam and compromising between the thermal effects of the two materials. These difficulties need to be resolved for the best performance and durability [120].

As shown in Fig. 15, environmental conditions, including Climate, Air Temperature, Humidity, and Dust Deposition, have significant effects on the durability of radiative paints and metal foam fins. In contrast, material properties, such as Reflectivity, Corrosion Resistance, and Porosity, govern their durability and performance. It effectively captures the correlation among these parameters. It provides a general sense of how environmental conditions, along with material properties, influence the long-term performance of passive cooling devices for PV applications. Metal foam fins and radiative paints are emerging techniques to improve PV panel performance, but their durability under real-world conditions needs to be investigated. Their combined use shows promise in extending the lifespan of PV panels, but material compatibility, thermal management, and cost remain challenges to their widespread adoption.

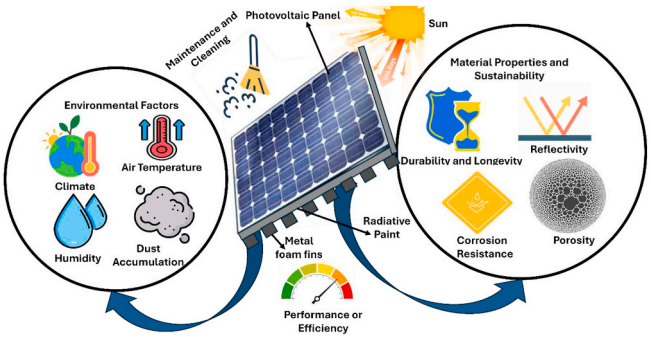


Fig. 15. Key factors influencing the durability, performance, and sustainability of passive cooling systems in PV panels.

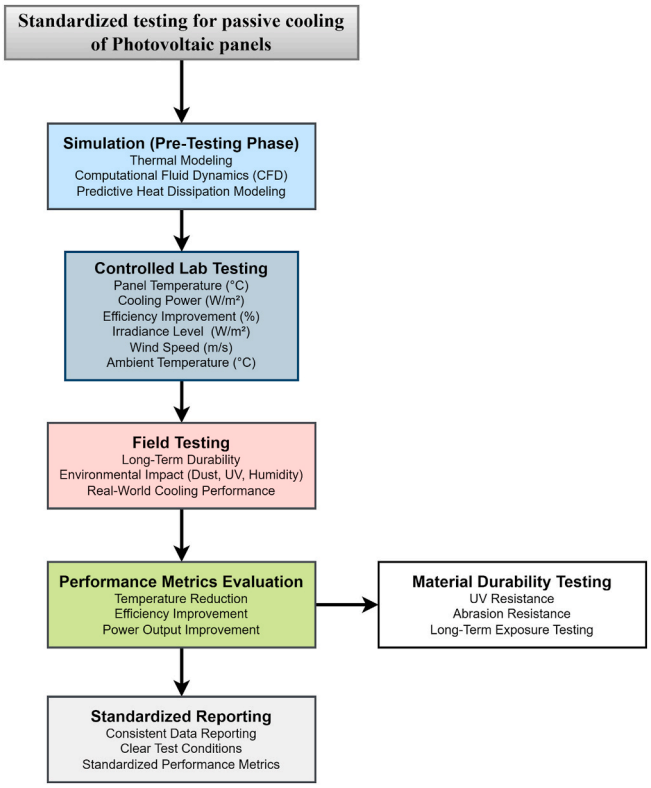


Fig. 16. Standardized testing framework for passive cooling of PV panels.

4.4. Standardized testing protocols

Standardized testing for passive cooling of PV panels is crucial to assess the effectiveness and reliability of various cooling methods. As illustrated in Fig. 16, this process follows a structured approach comprising multiple phases that ensure consistent, reproducible data across different methods and environmental conditions. The key phases

Table 6
Quantitative comparison of climatic adaptability for radiative coatings for PV panels.

Coating Type	UV Aging Resistance (h)	Reflectivity Retention (%)	Emissivity Stability (%)	Dust Adhesion Impact (% Loss)	Humidity Exposure Tolerance (%)	Reference (s)
Breathable weather-adaptive radiative coating (particle self-assembled)	960	≈94 (retained)	93 (stable)	≈0 (negligible)	100	[188]
Dynamically omniphobic and highly transparent polyurethane coating	1000	≈92 (retained)	–	≈2	100	[189]
SiO_2 hollow microsphere composite film	1000	≈93 (retained)	≈95 (stable)	≈3	100	[190]
TiO_2/PDMS radiative cooling coating (tropical-tested)	500 (PDMS-based, weather-resistant)	≈88 (retained)	≈90 (stable)	≈5 (minor surface impact under dust)	80 (performance reduced in high humidity, >80 % RH)	[191]
SiO_2 particle integrated superhydrophobic thermal insulating hollow nanofiber film	1000 (UV aging, no degradation)	≈95 (retained after aging)	≈96 (stable)	≈0 (negligible due to superhydrophobicity)	100 (maintained under high humidity)	[192]

include simulation (pre-testing phase), controlled lab testing, field testing, performance metrics evaluation, material durability testing, and standardized reporting.

Simulation (Pre-Testing Phase): Before performing physical tests, simulation models such as thermal modelling, CFD, and predictive heat-dissipation modelling are utilized. These models allow optimizing cooling configurations by simulating different designs, such as varying fin numbers, and assessing their impact on heat dissipation efficiency. By employing multi-objective optimization techniques, these models balance different cooling requirements, such as minimizing panel temperature while maximizing efficiency and reducing pressure drops [194]. This stage sets the groundwork for controlled lab testing and further experimentation.

Controlled Lab Testing: In this phase, PV panels are tested under controlled environmental conditions, measuring key parameters such as panel temperature (°C), cooling power (W/m^2), efficiency improvement (%), irradiance level (W/m^2), wind speed (m/s), and ambient temperature (°C). The goal is to assess how the cooling system reduces panel temperatures, improves electrical efficiency, and handles varying environmental conditions, such as sunlight intensity and ambient temperature. The performance metrics from this stage directly feed into subsequent evaluation processes [195].

Field Testing: Real-world testing is essential for evaluating how cooling methods perform in dynamic, unregulated environments. This includes assessing the durability of the cooling system under the effects of long-term environmental factors such as dust accumulation, UV exposure, and humidity. These factors significantly impact the real-world performance and longevity of PV panels, as dust can reduce the panel's efficiency by blocking sunlight, while UV radiation and humidity can degrade materials and reduce cooling effectiveness [196].

Performance Metrics Evaluation: Performance evaluation assesses the effectiveness of the cooling method by measuring temperature reduction, efficiency improvements, and power output enhancements. Lowering the panel temperature directly correlates with increased efficiency and power output, as cooler panels exhibit reduced electrical losses. By incorporating detailed simulation models and comparing results with experimental data, the cooling methods can be optimized further to improve performance [197].

Material Durability Testing: The durability of materials used in passive cooling solutions is also assessed to determine their long-term viability. Materials such as PCMs, heat pipes, and polymeric coatings are tested for their resistance to UV radiation, abrasion, and prolonged exposure to environmental conditions. Studies have shown that PCMs and heat pipes significantly improve PV performance, while polymeric coatings enhance the system's overall durability [198,199]. The material durability testing phase ensures that the cooling system maintains its performance over time, especially under extreme conditions.

Standardized Reporting: Finally, consistent data reporting is crucial for validating experimental results and ensuring meaningful comparisons across studies. Clear reporting standards are established for each testing phase, including controlled lab conditions, environmental factors, and performance metrics. By adhering to standardized reporting practices, researchers can ensure that their findings are reproducible and can be built upon in future studies [200].

By following these standardized testing protocols, researchers and industry professionals can ensure that passive cooling solutions for PV panels are not only effective in laboratory settings but also durable and efficient under real-world conditions. This approach provides a comprehensive evaluation of cooling methods, improving the performance and longevity of solar energy systems.

5. Challenges and scientific gaps

Despite the significant advancements in passive cooling technologies for PV panels, several challenges persist that hinder their widespread adoption and effectiveness. These challenges encompass issues related to material performance, environmental variability, and the absence of standardized testing protocols. As demand for more efficient, cost-effective cooling solutions grows, addressing these barriers is critical to ensuring the reliability and longevity of PV systems. Fig. 17 below illustrates the key scientific gaps and challenges in passive cooling for PV panels, providing a visual representation of areas requiring further research and development. These include critical issues such as the long-term durability of radiative paints, modelling and characterization of metal foam properties, environmental impacts on cooling performance, lack of standardization in performance evaluation, scalability and cost-effectiveness, and the need for long-term field validation. Identifying and addressing these gaps is essential to advancing the field of passive cooling in PV applications and ensuring their sustainable, widespread use.

5.1. Long-term durability of radiative paints

The long-term durability of radiative paints for PV applications remains a significant challenge. Radiative paints degrade due to several environmental factors, including UV radiation, heat, humidity, and atmospheric pollutants. These factors cause the paint to lose its reflectivity and, consequently, its cooling efficiency. While degradation mechanisms are well understood to some extent, the specific rates and pathways under different environmental conditions remain poorly understood. For instance, UV radiation causes paint to yellow, which negatively affects its reflective properties and reduces its cooling performance [50,201]. Humidity can weaken paint adhesion to the surface, leading to peeling or cracking. Additionally, airborne particles, dust, and surface contamination can accumulate on the paint, reducing its reflectivity and cooling efficiency [202].

The correlation between accelerated weathering tests and natural weathering is often weak. Accelerated tests, which simulate years of environmental exposure in a shorter period, do not always align with the degradation observed under real-world conditions. This discrepancy makes it difficult to predict the long-term performance of radiative paints. There is a need for more realistic weathering simulation methods that better replicate natural environmental conditions. Until this gap is addressed, it will remain challenging to predict the long-term performance and durability of radiative paints accurately [203].



Fig. 17. Challenges and scientific gaps in passive cooling for PV panels.

Furthermore, different radiative paints exhibit varied responses to weathering. Some materials show significant yellowing, cracking, and peeling, which directly affect their cooling performance. Understanding the material-specific degradation pathways is essential for improving the longevity of these paints. Each paint type may degrade in different ways depending on its chemical composition and formulation. To improve durability, it is crucial to investigate these material-specific responses more thoroughly and identify strategies to mitigate these effects [69].

To address these gaps, several solutions can be explored through material innovation. One such solution involves incorporating advanced materials such as photocatalysts and self-healing compounds into the formulations of radiative paints. These advanced materials could improve the paint's resistance to environmental degradation and enhance its ability to maintain its cooling performance over time [202]. Additionally, the use of eco-friendly, water-based formulations with optimized particle size distributions could increase both the durability and environmental sustainability of the paints [204].

Surface treatments also play a crucial role in improving the durability of radiative paints. For instance, silicone-based binders have shown promising results in enhancing the paint's resistance to UV radiation and contamination. These treatments could significantly extend the lifespan of radiative paints by improving their ability to resist environmental stressors. Developing coatings with self-cleaning properties, such as those incorporating hydrophobic or photocatalytic surfaces, is another promising direction. These treatments can help maintain the paint's reflectivity and cooling performance by preventing the accumulation of dirt and pollutants on the paint's surface [202].

In addition to material innovation, developing standardized testing protocols is essential to assess the long-term durability and performance of radiative paints. A combination of natural weathering tests and improved accelerated weathering protocols would provide more reliable insights into the materials' long-term performance. While natural weathering tests offer realistic data, accelerated weathering tests can yield results more quickly. However, the correlation between the two must be improved to ensure that accelerated tests accurately represent real-world conditions. Advanced analytical techniques such as Fourier Transform Infrared Spectroscopy (FTIR), X-ray Photoelectron Spectroscopy (XPS), and chemiluminescence can help monitor subtle changes in paint properties over time and provide more accurate predictions of long-term performance [205].

Moreover, standardizing testing methods is crucial to ensuring reproducibility and consistency across studies. Existing standards, such as ISO 12944 for paint durability and IEC 60068-2 for environmental testing, can serve as a foundation. However, these standards need to be adapted to suit the specific requirements of radiative paints in PV applications. Inter-laboratory studies and field tests should be conducted to validate these methods and ensure that they apply to the diverse environmental conditions in which PV panels operate [206].

By addressing gaps in understanding and material innovation, as well as by developing robust, standardized testing protocols, the long-term durability and performance of radiative paints can be significantly improved. This will ensure that they remain effective for various applications and contribute to the overall efficiency of PV systems.

5.2. Modelling and characterization of metal foam properties

Metal foam fins, utilized in passive cooling systems for PV panels, present significant modelling and characterization challenges due to their complex geometries and varied material properties. The intricate internal structure of metal foams makes it challenging to model accurately through conventional techniques. Advanced computational methods, such as micro-computed tomography (μ-CT) scans, are increasingly used to generate precise 3D representations of these structures, enabling more accurate predictions of heat transfer performance. Despite these advancements, the complexity of metal foam geometry remains a challenge for simulations, leading to inaccuracies and high computational costs [207]. Additionally, metal foams exhibit variable porosity and pore density, which directly affect thermal performance but are often simplified or assumed constant in current models, leading to less reliable predictions [208].

Key characteristics, including porosity, pore size, and material composition, strongly influence the thermal behavior of metal foams. High porosity enhances the surface area, which is beneficial for heat transfer, but it also increases pressure drop, complicating the optimization process. Research has shown that optimal porosity levels, typically around 0.85, balance heat transfer and pressure drop more effectively [105,209]. Furthermore, pore size plays a critical role in enhancing convective heat transfer, as smaller pores generally increase the surface area for heat exchange but may exacerbate pressure drop. Aluminium and copper foams are widely studied for their thermal conductivity, with aluminium being more cost-effective, while copper foams provide superior performance but at a higher cost [15,210]. These material-specific characteristics need to be considered when designing metal foam fins for PV applications.

To improve the accuracy of modelling metal foam fins, the development of more comprehensive and dynamic models is essential. Incorporating varying porosity and pore density into simulations can enhance their predictive accuracy, ensuring better alignment with real-world performance. Additionally, accounting for the effects of compression on foam behavior is crucial, as compressed foams exhibit different thermal properties compared to uncompressed ones [211]. Experimental validation is also necessary to ensure the reliability of numerical models, with consistent measurement techniques, such as the ASTM 2-bar method for thermal conductivity, being vital for standardization. By refining these models and incorporating detailed experimental data, the effectiveness of metal foam fins in PV cooling applications can be significantly improved, offering a promising avenue for enhancing PV efficiency [102,212].

5.3. Environmental impacts on cooling performance

Environmental factors, such as dust accumulation, humidity, and pollution, significantly impact the cooling efficiency of passive cooling technologies used in PV panels. Dust accumulation on radiative paints and metal foam fins can reduce their cooling performance by obstructing heat dissipation and blocking sunlight, which is critical for passive cooling systems [202,213]. High humidity levels also negatively affect the performance of these cooling materials. For instance, in humid tropical environments, cooling materials such as porous polymer coatings exhibit a significant decrease in cooling efficiency, as they lose their ability to maintain sub-ambient cooling under high humidity [191]. Moreover, air pollution, outstanding particulate matter, can further degrade the efficiency of these passive cooling systems by accumulating on surfaces and hindering their thermal performance [214]. These challenges highlight the importance of accounting for environmental factors in the design and implementation of passive cooling technologies for PV systems.

Despite these recognized challenges, significant gaps remain in understanding the long-term effects of environmental factors on cooling materials. Many current studies focus on short-term impacts, neglecting the sustained performance degradation caused by dust, humidity, and pollutants over extended periods [202,215]. Additionally, the durability of cooling materials under varying environmental conditions, including their resistance to soiling, UV radiation, and atmospheric pollutants, remains underexplored [216]. To address these gaps, future research should develop more comprehensive performance models that integrate environmental stressors and conduct long-term environmental testing. Enhanced material development, such as coatings with self-cleaning and hydrophobic properties, could improve resilience to these environmental challenges and ensure the long-term effectiveness of passive cooling technologies in PV applications.

5.4. Lack of standardization in performance evaluation

One significant challenge in evaluating passive cooling systems for PV panels is the lack of standardized performance metrics. The performance of passive cooling technologies, such as radiative paints and metal foam fins, is often evaluated under varying environmental conditions, such as different temperatures, humidity levels, and dust accumulation, making direct comparisons between studies difficult. The lack of universally accepted metrics, such as cooling efficiency, heat dissipation rate, and durability, leads to inconsistent performance evaluations, making it difficult to assess the actual effectiveness of these systems across different research contexts [217]. Additionally, the complex interactions between PV panels and cooling technologies, including factors like panel orientation and installation conditions, make it challenging to develop standardized metrics that accurately reflect performance in real-world scenarios [218]. Moreover, variations in measurement techniques and equipment further exacerbate inconsistent results, as different studies may use various methods to assess key parameters such as temperature reduction or heat flux [219].

To address these challenges, it is essential to develop standardized testing protocols that account for the diverse environmental conditions in which passive cooling systems operate. Establishing a set of universal testing conditions, including standardized temperature, humidity, and solar irradiance levels, would allow for more meaningful comparisons between different passive cooling technologies [106]. Furthermore, defining key performance metrics and ensuring consistent measurement techniques across studies will help reduce discrepancies in results and improve the reproducibility of findings [220]. Initiatives like the European Partnership on Metrology's PaRaMetriC project, which aims to define figures of merit for passive cooling materials, provide a valuable framework for standardizing the evaluation of radiative paints and metal foam fins [216]. By implementing these standardized protocols and measurement guidelines, the scientific community can ensure more reliable, consistent, and reproducible performance data, facilitating broader adoption of passive cooling solutions for PV applications.

5.5. Scalability and cost-effectiveness of passive cooling solutions

The scalability of passive cooling solutions for PV installations faces challenges related to both radiative paints and metal foam fins. For radiative paints, challenges arise from the balance between cooling efficiency and material transparency, with more cost-effective options like PVB being scalable but less effective than high-performance coatings like PDMS [120]. Additionally, scaling up manufacturing to meet the demands of large PV installations involves overcoming issues like manufacturing complexity and the high costs of selective emitters, which can hinder cost-effectiveness in large deployments [221]. For metal foam fins, the cost of production, primarily through casting methods, remains high due to the need for precise control over the foam structure [76,145]. Scaling these methods for large-scale manufacturing, while maintaining performance standards, presents a significant challenge, particularly with the use of high-conductivity materials like beryllium alloys or aluminium nitrides that are cost-prohibitive at scale [222].

To address these scalability challenges, several strategies can optimize both radiative paints and metal foam fins for cost-effective large-scale applications. For radiative paints, reducing material costs by using commercially available polymers and incorporating cost-effective additives, such as glass bubbles, can enhance cooling performance without significantly increasing costs [14]. In metal foam fins, innovations such as die casting and additive manufacturing are key to lowering costs while improving thermal conductivity and performance [223]. Economies of scale also play a crucial role in reducing costs as PV installations grow. By fostering large-scale PV deployment, localized manufacturing and supply chains can reduce the overall cost of passive cooling technologies, making them more accessible and affordable for widespread adoption in diverse markets [224].

5.6. Long-term field validation and real-world performance

Long-term field validation of passive cooling technologies, such as radiative paints and metal foam fins, for PV applications faces several significant gaps. One key challenge is the environmental impact of material degradation over time. Passive cooling systems are exposed to various environmental factors, such as UV radiation, dust, humidity, and temperature fluctuations, which can degrade their performance. For instance, radiative paints suffer from soiling, where dust accumulation blocks the cooling effect, and their effectiveness is also diminished by UV weathering, making their long-term performance less reliable in real-world settings [202,225]. Metal foam fins, although promising in terms of heat dissipation, face challenges related to mechanical wear and corrosion under prolonged exposure to outdoor elements [15]. Furthermore, there is a lack of standardized testing protocols that can consistently evaluate the long-term performance of these materials across various environmental conditions. This inconsistency makes it difficult to compare results across different studies, which undermines the development of universally accepted performance metrics for passive cooling systems [219,226].

To incorporate long-term environmental conditions into performance evaluations, several strategies need to be adopted. One approach is to develop comprehensive testing protocols that simulate long-term exposure to environmental stressors, such as UV radiation, dust, and humidity. This can be done using environmental simulation chambers that replicate diverse real-world conditions, providing more accurate data on material performance [227]. Another strategy involves enhancing the materials used in passive cooling systems to improve their durability. For instance, UV-resistant and hydrophobic materials can prevent degradation caused by sunlight and moisture, thereby maintaining cooling efficiency over extended periods [202]. Additionally, field studies that collect long-term performance data from real-world installations are essential for understanding the practical effectiveness of passive cooling systems. These studies can serve as a feedback loop, where insights gained from real-world performance inform material design improvements and help refine existing cooling systems [15,228]. Incorporating such real-world data into the development process can lead to more resilient and efficient systems that are better suited for large-scale adoption in PV applications.

6. Proposed solutions and future directions

The development of passive cooling technologies for PV panels is crucial for enhancing their efficiency and longevity, particularly in regions with high solar irradiance. This section outlines the primary cooling solutions for PV panels, focusing on the integration of metal-foam fins and radiative cooling systems. These technologies aim to improve thermal management, reduce panel temperatures, and enhance the overall performance of solar systems, ensuring their long-term durability. Fig. 18 below provides a visual overview of these proposed cooling solutions, illustrating the designs and cooling mechanisms of both systems: metal foam fins for convective heat dissipation and radiative paints for enhanced infrared emission. This highlights how these systems can be integrated into PV panels to optimize heat regulation, improve energy efficiency, and extend panel lifespan.

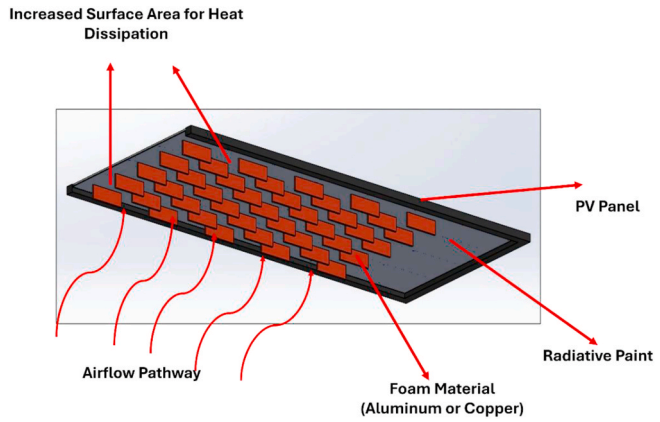


Fig. 18. Proposed passive cooling solutions for PV panels.

6.1. *Material innovations for enhanced durability*

The development of advanced materials is essential to enhancing the efficiency and longevity of passive cooling technologies in PV panels, particularly in regions with harsh environmental conditions such as UV radiation, humidity, and dust accumulation. Metal foam fins and radiative paints are two primary technologies being explored to improve thermal management in PV systems. These materials must withstand prolonged exposure to environmental elements while optimizing thermal regulation and improving panel performance. Metal foam fins, made from aluminium and copper, are favored for their high thermal conductivity, light weight, and porous structure. For instance, aluminium foam is known for its ability to dissipate heat efficiently, and when engineered with optimal porosity and pore density, it significantly improves cooling efficiency. Studies by Tan et al. [229] and Cheong Tan et al. [230] demonstrate that aluminium foam enhances thermal management in PV panels by increasing the surface area for heat exchange, thereby boosting cooling efficiency. Copper foam, with its higher thermal conductivity, is particularly effective for high-performance PV applications that require rapid heat dissipation. Research by Hasan and Farhan [15] showed that open-cell copper foam fins effectively reduce the operating temperature of PV panels, leading to improved efficiency and power output.

The convective cooling benefits of metal foam fins arise from their porous structure, which allows air to flow through the fins and increases the surface area available for heat exchange. This facilitates more efficient heat dissipation, helping maintain uniform temperature and improving the overall performance of PV panels. Furthermore, integrating (PCMs) with metal foam fins has the potential to enhance cooling by utilizing PCM's latent heat storage capacity, extending the cooling effect over a longer period. Rahmanian-Koushkaki et al. [231] demonstrated that combining metal foam fins with PCMs significantly improves cooling efficiency in concentrated photovoltaics (CPV) systems, providing sustained cooling effects.

In addition to metal foam fins, radiative paints are also crucial for improving the thermal management of PV systems. These paints, which often contain materials such as TiO_2 and SiO_2, work by reflecting solar radiation and emitting infrared radiation, thereby lowering the temperature of PV panels. Li et al. [232] showed that TiO_2 nanoparticles increase reflectivity and enhance the mechanical resistance of the paint, thereby improving durability and efficiency under harsh environmental conditions. Furthermore, incorporating calcium phosphate and glass bubbles into radiative paints enhances solar reflectance and improves overall durability, ensuring the paint remains effective even under long-term exposure to the elements [233].

Both metal foam fins and radiative paints face environmental degradation challenges, such as corrosion and UV exposure. To enhance the durability of metal foam fins, surface treatments such as nickel plating, gold plating, and silica primers are applied to improve corrosion resistance and maintain mechanical stability under extreme conditions. Oriňáková et al. [234] demonstrated that nickel coatings significantly improve the corrosion resistance of metal foams, ensuring their long-term performance in harsh environments. Radiative paints are particularly susceptible to UV degradation, which can cause oxidation, yellowing, and binder deterioration. To mitigate these issues, strategies such as the use of high-bandgap materials, UV stabilizers, and protective coatings are being explored. Lio et al. [130] highlighted the effectiveness of UV stabilizers in maintaining the reflectivity and aesthetic quality of radiative paints over time, helping to prolong their durability.

Both metal foam fins and radiative paints play vital roles in improving the thermal management of PV panels. Metal foam fins primarily enhance convective cooling, while radiative paints reduce the heat absorbed by the panels through infrared emission. While both technologies have been shown to improve cooling efficiency and panel performance individually, there is currently no literature reporting their combined application in PV systems. To ensure the long-term durability of each technology, both metal foam fins and radiative paints require surface treatments and protective coatings to mitigate corrosion, UV degradation, and other environmental stressors. As material innovations continue, these cooling technologies hold great potential to improve the efficiency of solar energy and the sustainability of PV systems. However, further research is needed to explore the synergistic effects of combining these technologies to optimize the thermal performance of the PV panel.

6.2. *Improving modelling, testing, and characterization techniques*

Current modelling techniques for predicting the performance of passive cooling systems in PV panels, such as metal foam fins and radiative paints, face several limitations, including the failure to account for real-world variables such as wind velocity, temperature fluctuations, and varying solar irradiance. These variables significantly affect the cooling performance of PV systems, leading to discrepancies between theoretical models and actual performance. For instance, a study by Elavarasan et al. [235] demonstrated that simplified models used in numerical simulations of PV systems fail to capture the impact of dynamic environmental changes, leading to a performance prediction error of up to 9 % under uncontrolled conditions. To overcome these challenges, future research should focus on incorporating real-world environmental variables into computational models. Techniques such as CFD combined with Monte Carlo Ray-Tracing can be employed to accurately simulate the complex thermal and radiative behavior of cooling systems under various environmental conditions. Additionally, integrating machine learning algorithms with these models can enhance predictive accuracy by leveraging real-time data, making the models more adaptive to fluctuating conditions and improving their reliability.

Alongside advancements in modelling, experimental validation is crucial to ensure that theoretical predictions align with real-world performance. Experimental testing of passive cooling systems in PV panels under a range of environmental conditions, including varying solar irradiance, humidity, and temperature, is necessary to calibrate and validate the models. As highlighted by Novak et al. [11], real-time dynamic weather changes significantly influence cooling efficiency, and testing under controlled conditions is essential for accurate model validation. Moreover, to validate and characterize the effectiveness of passive cooling technologies, performance metrics such as temperature reduction and electrical efficiency should be measured in both prototype and full-scale PV installations. Prototypes enable evaluation of cooling efficiency and its impact on system performance, while long-term testing ensures the durability and sustainability of cooling technologies under continuous operation. Combining advanced computational techniques with robust experimental testing will enhance the accuracy, scalability, and real-world applicability of passive cooling systems for PV applications.

6.3. Environmental adaptability, optimization, and integration with hybrid systems

Optimizing passive cooling technologies, such as metal foam fins and radiative paints, for varying environmental conditions, including humid, dusty, or arid climates, requires addressing specific challenges that affect their efficiency and durability. In humid climates, high moisture levels can reduce the cooling efficiency of radiative paints by reducing their thermal emissivity. To mitigate this, hydrophobic coatings can be applied to prevent water accumulation and maintain the effectiveness of these systems [236]. In dusty environments, particle accumulation on cooling surfaces significantly reduces their reflective and emissive properties, thereby reducing cooling performance. To counter this, self-cleaning coatings or hydrophobic surfaces can be used to keep dust from sticking to the panels [237]. In arid climates, where moisture is scarce, evaporative cooling methods become less effective. However, hybrid systems that combine passive cooling techniques with active cooling solutions, like Photovoltaic-thermal (PV/T) systems, can optimize cooling performance. These hybrid systems leverage both radiative cooling and active thermal management, which can be particularly beneficial in hot and dry regions, ensuring efficient heat dissipation and improved energy output. By integrating self-cleaning technologies, hydrophobic coatings, and hybrid cooling systems, passive cooling solutions can be optimized to perform effectively under varying environmental conditions, ensuring the longevity and reliability of PV panels in diverse climates.

The integration of passive cooling systems with hybrid technologies can significantly enhance the efficiency and sustainability of PV panels. Hybrid systems, such as PV/T systems, combine passive cooling strategies, such as metal foam fins or radiative paints, with active cooling methods, thereby improving thermal management and enhancing overall system performance. By cooling both the PV cells and the associated thermal system, these hybrid systems increase the energy output of PV installations while simultaneously providing sound thermal energy, such as hot water. Economically, hybrid systems offer higher returns on investment by producing both electricity and thermal energy, reducing energy costs and improving system performance. Environmentally, these systems contribute to sustainability by minimizing the need for additional cooling systems and utilizing waste heat, thereby reducing greenhouse gas emissions. Furthermore, machine learning can optimize hybrid system performance by analyzing real-time environmental data and adjusting the cooling strategy accordingly, ensuring optimal efficiency [238]. However, scaling the integration of passive cooling technologies with hybrid systems for large-scale PV installations poses challenges, including system complexity, cost, and the need for robust testing protocols across diverse climates. Standardized testing methods that incorporate local climate data and real-world variables will be critical for accurately assessing the performance of these systems and ensuring their viability for commercial PV applications.

6.4. Scalability, cost-reduction, and real-world validation

Scaling up passive cooling technologies, such as metal foam fins and radiative paints, for large-scale PV systems poses several challenges, including material complexity, manufacturing costs, and environmental durability. Radiative paints must exhibit high solar reflectivity and strong emissivity within the atmospheric window to ensure adequate cooling, yet these properties must be balanced with mechanical robustness and transparency [120,239]. Meanwhile, the fabrication of metal foam fins via techniques such as investment casting or additive manufacturing requires precise control of porosity and structure, increasing production costs and limiting scalability [240]. However, advances in materials like PVB coatings and optimization techniques such as response surface design and 3D printing offer pathways to reduce cost while preserving thermal performance [241].

Real-world validation is essential to confirm the feasibility of these technologies across diverse climates. Pilot projects and field studies conducted in regions with varying solar irradiance, humidity, and wind enable performance benchmarking under actual operating conditions. Empirical evidence has shown that integrating copper metal foam fins can reduce PV surface temperatures by up to 13.4 %, improving power output by nearly 5 %, while radiative coatings offer similar enhancements with minimal maintenance requirements [15,98]. Combining these results with CFD simulations and cost modelling enables the calculation of financial indicators such as Levelized Cost of Electricity (LCOE), payback period, and Net Present Value (NPV). Studies report LCOE values in the range of 0.10–0.12 €/kWh and payback periods between 5 and 10 years, supporting the economic case for wide-scale deployment of passive cooling in solar infrastructure [41,242].

6.5. Potential impact and benefits

Passive cooling techniques, such as radiative paints and metal foam fins, also demonstrated significant potential for lowering the operating temperature of PV panels, particularly in regions of high irradiation. For example, open-cell copper metal foam fins decreased panel temperature by 8.4 % and enhanced power output by 4.9 % during a laboratory test in Baghdad, Iraq [15]. Temperature reduction with significant efficiency enhancement during solar noon was also verified in another study using aluminium fins [243].

By minimizing thermal stress, these passive systems improve electrical efficiency and increase the working lifespan of PV modules. Hasanuzzaman et al. [244] presented in a study that passive cooling can increase the efficiency of PV by as much as 15.5 %, with each 1 °C decrease averting a 0.5 % loss in efficiency. Prolonged lower operating temperatures also diminish material degradation, allowing longer PV system life [245]. In addition, the use of metal foam fins and controlled water flow was found to enhance the thermal efficiency by 4.7 % [246], whereas corrugated fins with denser pores obtained up to 86 % thermal-hydraulic efficiencies [93].

These improvements have not just technical, but also environmental advantages. PCMs, for instance, can reduce surface temperature by 24 °C, or by the equivalent 200–300 kWh of saved electricity annually [247]. Nocturnal and diurnal radiative cooling techniques allow heat dissipation without energy input and are therefore more sustainable [248]. However, passive techniques such as natural convection and reflective coating have also been proven to enhance the performance of solar panels by reducing the utilization of active cooling systems [249]. Overall, integrating these technologies enables solar PV systems to achieve greater efficiency, longer lifespan, and a smaller environmental footprint in practical applications.

7. Conclusion

This review critically examines two new passive cooling technologies for PV panels, radiative paints, and metal foam fins, and their thermal mechanisms, material compositions, structural configurations, and performance metrics. The radiative paints provide sub-ambient cooling by reflecting and emitting backside solar radiation in the atmospheric transparency window, whereas the metal foam fins provide high convective heat dissipation owing to their porous structure and high surface area. Both techniques have shown tremendous potential to minimize PV module temperature, enhance electrical efficiency, and prolong panel lifespan without additional energy consumption or mechanical complications.

The comparative analysis indicated that material selection, porosity, spectral selectivity, and environmental conditions are the predominant factors governing the cooling performance of these technologies. However, several challenges persist, including degradation of coating materials under environmental exposure, limited standardization of testing protocols, and incomplete understanding of long-term field performance.

A significant contribution of this review is the proposal of a hybrid passive cooling system that integrates radiative paints with metal foam fins. This novel concept combines radiative and convective cooling pathways to achieve enhanced thermal regulation. While theoretical justification supports the potential for synergistic performance, no experimental studies currently validate this combined approach. This represents a promising direction for future research.

To fully harness the benefits of passive cooling in PV applications, further studies are needed to optimize material properties and system configurations, investigate long-term durability under real-world conditions, and evaluate techno-economic feasibility. The development of scalable, low-maintenance, and climate-resilient passive cooling systems will be vital for improving the energy yield and reliability of PV installations, especially in off-grid and high-temperature environments. By bridging current knowledge gaps and proposing a forward-looking solution, this review contributes a foundational step toward sustainable and efficient solar energy technologies.

CRediT authorship contribution statement

Kehinde Temitope Alao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kamaruzzaman Sopian:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition. **Syed Ihtsham Ul Haq Gilani:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition. **Jing Ee Yit:** Writing – review & editing, Supervision. **Shuib Husin:** Writing – review & editing. **Taiwo Onaopemipo Alao:** Writing – review & editing. **Zeshan Aslam:** Writing – review & editing. **Hussein A. Kazem:** Writing – review & editing, Supervision.

Declaration of competing interest

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Data availability

No data was used for the research described in the article.

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